

Received 8 July 2025, accepted 20 July 2025, date of publication 25 July 2025, date of current version 31 July 2025.

Digital Object Identifier 10.1109/ACCESS.2025.3592668

RESEARCH ARTICLE

Comparing Machine Learning-Based Crime Hotspots Versus Police Districts: What's the Best Approach for Crime Forecasting?

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This work was supported in part by the “Piano Nazionale di Ripresa e Resilienza (PNRR) Italian Ministry of University and Research (MUR) Project” under Grant PE0000013-FAIR; in part by the “ICSC National Centre for HPC, Big Data, and Quantum Computing” with the NextGenerationEU Program under Grant CN00000013; in part by European Union-NextGenerationEU-National Recovery and Resilience Plan, PNRR-Project: “SoBigData.it-Strengthening the Italian RI for Social Mining and Big Data Analytics” under Grant Prot. IR0000013-n. 3264, 28/12/2021; and in part by the Research Project “INSIDER: INtelligent SerVice Deployment for advanced cloud-Edge integration” grant funded by MUR within the PRIN 2022 Program and European Union-Next Generation EU under Grant 2022WWSRR and Grant CUP H53D23003670006.

ABSTRACT Criminal activity poses a significant challenge in urban environments, impacting public safety, economic stability, and overall quality of life. As a result, the efficient allocation of public security resources based on spatio-temporal crime prediction models has become a critical concern for urban management. To this purpose, many crime forecasting approaches first split city territories into partitions based on crime rates and trends, with each partition reflecting criminal dynamics of its specific area. Then, crime forecasting models are extracted for each area, to monitor and predict how crime rates evolve over time within each partition. However, traditional spatial partitioning approaches, which divide cities into predefined police districts based on geographic and operational considerations, often fail to account for variations in crime patterns. In contrast, machine learning-based approaches could dynamically adapt to areas with differing crime frequencies and densities, making them particularly effective in cities characterized by diverse population distributions and crime activity levels. This study examines the impact of various partitioning techniques on crime forecasting performance, comparing the traditional static division of the city into police districts with machine learning approaches, specifically density clustering algorithms, for detecting crime hotspots. The experimental evaluation, conducted on two real-world case studies, i.e. Chicago and Los Angeles crime data, demonstrates the effectiveness of density-based clustering in identifying multi-density crime hotspots. Compared to traditional police district partitioning, these data-driven methods offer significant advantages in improving crime forecasting accuracy across urban environments.

INDEX TERMS Crime forecasting, multi-density clustering, crime hotspots, regression analysis, smart cities, crime data mining, smart cities.

I. INTRODUCTION

Criminal activity poses a significant challenge in urban environments, impacting public safety, economic stability, and sustainable development, and can lower residents' quality

The associate editor coordinating the review of this manuscript and approving it for publication was Tiago Cruz¹.

of life and well-being. As a result, developing methods for the efficient allocation and use of limited public security resources has become a critical concern for policymakers and urban management departments [5], [11], [42], [43]. To this purpose, the deployment of pervasive urban sensor networks and the availability of ICT technologies in city areas is allowing public institutions and law enforcement agencies

to gather extensive crime-related data with spatial and temporal dimensions. Analyzing this massive amount of data through machine learning enables the discovery of valuable descriptive and predictive models, which can be leveraged for accurate spatio-temporal crime forecasting. By extracting these predictive, data-driven insights, police departments and public security agencies can optimize resource utilization and implement more targeted crime prevention strategies.

Several studies proposed in the crime forecasting literature show that criminal events are not uniformly distributed across urban areas [5], [10], [14], [16], [27], [42], [43]. In fact, crime rates vary significantly depending on geographic factors, with certain neighborhoods exhibiting higher risk levels than others. Moreover, crime rates exhibit temporal variations, including peaks, troughs, and seasonal trends that follow distinct patterns over days of the week, months, or even years. For such a reason, many spatio-temporal crime prediction approaches first split city territories into partitions based on crime rates and trends, where each partition reflects specific criminal dynamics of its specific area. Then, crime forecasting models are extracted for each area, to monitor and predict how crime rates evolve over time within each partition [6], [10], [14], [25]. These insights enable police departments to optimize resource distribution across urban areas, allowing for the strategic deployment of officers to high-risk zones or reallocating personnel from areas with expected declines in criminal activity. In this way, police can prevent or respond to crimes more effectively.

A first classic method to split the urban territory for crime prevention is based on a static division of the city in *police districts*. Here, the city is divided into predefined partitions, where boundaries are designated by law enforcement on the basis of neighborhood borders, major streets, population density, operational demands, costs, and other topological aspects of the urban environment [30]. This partitioning approach aligns police coverage with practical and geographic characteristics, providing a foundational framework for patrols and resource allocation. This technique, increasingly aided in recent years by Geographic Information Systems (GIS) and mathematical models, typically relies on the expertise and intuition of urban planners and law enforcement professionals for designing police districts. In some (very rare) cases, police district boundaries are adjusted and re-drawn to address demographic and structural changes within the city [41]. However, while expert insight can be valuable in initial districting, the traditional partitioning process may lag behind dynamic changes in demography, behavior patterns, and population dynamics [30].

A second way to partition urban areas to detect crime hotspots is based on *machine learning* algorithms, most of them based on *density-based clustering* techniques [9], [10], [15], [32], [34]. The underlying idea of these approaches is to apply clustering methodologies to automatically trace boundaries of city areas where crimes occur with higher densities and frequencies than other areas. In fact, as shown

in Figure 1 the density of population, traffic, or events in large cities can vary widely from one area to another area [16], which also makes the incidence of crime events extremely dissimilar in terms of density [14], [16], which is a key issue when correctly detecting the real crime hotspots from crime data. Recent research studies [13], [14], [16] demonstrate that *multi-density clustering* outperforms traditional methods in identifying hotspots within urban environments characterized by varying density levels, especially when crime event density varies significantly (which is very common in urban areas). This allows the discovery of *multi-density hotspots* (i.e., different regions having various densities [1], [2]), which are usually not properly detected by classic techniques unless the clusters (or hotspots) are clearly separated by sparse regions [19].

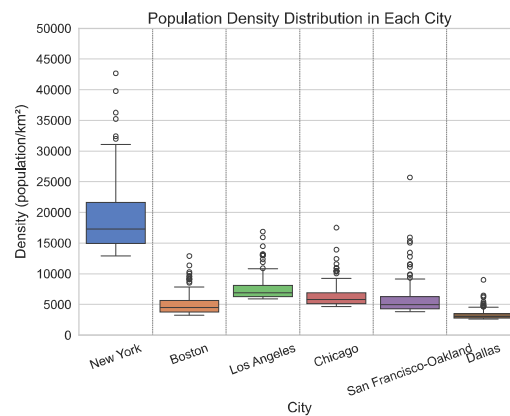


FIGURE 1. Intra-city and inter-city population densities in metropolitan urban areas. Population densities of the densest 200 cells for a given set of cities. Each cell has a 1 km^2 area.

Given the presented context, this paper presents a comparative study on hotspots detection in urban environments for crime forecasting. As the main contribution, the manuscript compares the results achieved when relying on a static division of the city in police districts and when hotspots are built by density-based clustering algorithms. Specifically, we evaluate three clustering techniques: HDBSCAN [8], CHD [16], and DBSCAN [19]. The experimental evaluation is conducted on two real-world crime datasets from the Chicago and Los Angeles Police Departments, both of which inherently characterized by points distributed with very different densities in the city area. Both cities are large urban environments with millions of recorded crime incidents, making them well-suited for evaluating the performance of the approaches. Moreover, such concrete scenarios are exploited to show the practical usefulness of density-based clustering algorithms in discovering multi-density urban hotspots in real urban cases.

The rest of the paper is structured as follows. Section II describes the most popular methodologies to partition the urban territory in crime hotspots, and the most representative projects in that field of research. Section III defines the problem and the goals of the approach. Section IV presents

a selection of the approaches we will exploit in this paper, for crime hotspots detection (including police district partitioning and density-based clustering) and crime forecasting, summarizing how they work. Sections V and VI provide the comparative experimental evaluations of the different approaches conducted on Chicago and Los Angeles datasets, respectively. Section VII offers a discussion of key findings, insights, and open questions that have emerged from this research study. Finally, Section VIII concludes the paper and plans future research works.

II. RELATED WORK

Hotspot detection and event forecasting have recently gained significant attention within the research community, for general urban data analysis and crime forecasting in particular. This section offers a brief review of the most notable studies in this domain.

The study in [12] presents an approach based on density-based clustering to group together sources and destinations of public transportation trajectories, e.g., taxi trajectories, in order to detect mobility patterns in cities. The work exploited the DBSCAN algorithm for clustering, and has been validated on a real-world dataset covering taxi trajectories of Beijing city. In the paper [6], cluster analysis was performed to group similar noise road events in order to characterize roads. The approach has been validated in a dataset comprehending noise events in the city of Milan (Italy). A similar approach has also been exploited in [31], which proposes a Waste-Volume management methodology for smart cities. The proposed methodology clusters together waste generator sites by means of statistical and density-based algorithms and then trains ARIMA- and neural network-based models for forecasting waste volumes in each cluster.

As pointed out by recent surveys [5], [21], data partitioning and hotspot detection, also referred to as spatiotemporal region discretization [21], are common fundamental tasks in analyzing urban events for forecasting and prediction purposes, in particular when analyzing crime datasets. For this purpose, the study proposed in [5] identifies a set of the most used techniques, which are Kernel Density Estimation [3], [23], [26], [38], Random Forests [4], [44] and density-based clustering algorithms [10], [18]. The research works described in [5], [14] claim also two results: density-based clustering algorithms perform better than others in detecting hotspots of spatiotemporal events, and clustering algorithms that are capable of detecting clusters having different densities perform even better when analyzing real-world data.

The work presented in [25] proposes a framework for cluster detection using k-means and crime forecasting by neural-fuzzy networks. The results on the UCI crimes dataset “Communities within the United States”, consisting of 43,420 crime events, are compared with other forecasting techniques such as Regression Tree and Gaussian Process Regression, where the proposed approaches achieved better

scores, in terms of RMSE and MAE with respect to the others. The paper [10] presented a framework based on events partitioning through DBSCAN and forecasting through SARIMA models. The approach has been validated by analyzing two large crime events datasets, set in Chicago and New York City, consisting of about 1,800K and 1,500K crime events, respectively, achieving up to 7.5 MAPE error in a one-year ahead prediction on seven-day aggregation windows. In [6], authors present an approach for crime events prediction, based on a combination of HDBSCAN for detecting crime hotspots and SARIMA models for forecasting, considering monthly time windows. The authors motivate the choice of the HDBSCAN algorithm as it can discover clusters having diverse densities w.r.t. DBSCAN. The forecasting approach has been exploited on the New York City crime dataset, considering monthly time windows, and the results have been critically compared to the paper [10]. The work presented in [14] explores the use of a multi-density clustering algorithm, purposely tailored for urban events partitioning, CHD [16], combined with SARIMA and LSTM models for forecasting urban events. The proposal has been validated against crime events of Chicago City (2001-2019), consisting of about 2,300K events. The achieved results were compared with other approaches, combining clustering algorithms such as DBSCAN, OPTICS, and HDBSCAN with SARIMA and LSTM for forecasting, showing that multi-density-based clustering algorithms performed better than others.

This latter aspect is also analyzed in [13], which studies how other clustering techniques, based on multi-density approaches, outperform classic approaches to discover urban hotspots. More specifically, the paper compares the DBSCAN, OPTICS-xi, HDBSCAN, and CHD algorithms against synthetic and real datasets, by selecting the best fitting algorithm parameters through a parameter sweeping approach. The analysis shows that the HDBSCAN and CHD algorithms are the most effective in detecting clusters in multi-density datasets, and that CHD performs better than HDBSCAN on the real-world dataset.

A systematic review of several works about crime hotspot detection and crime prediction is presented in [5]. In particular, the paper analyzes the impact of clustering techniques on the discovery of crime hotspots, and how time series analysis and deep learning techniques can be exploited for crime trend prediction. The review shows that ARIMA and LSTM models are the most used techniques for predicting crime trends in urban environments. The review also highlights the need, for comparison purposes, to exploit publicly available data to assess crime prediction results, and that the most widely exploited measurements for evaluating the effectiveness of the different approaches are MAE, MAPE, and RMSE, and suggests the use of relative performance indexes, such as MAPE, to simplify the comparison between different approaches.

With respect to the surveys in [5], [21], we want to highlight more recent and relevant works in crime

TABLE 1. Related work comparison table.

Publication	Goals	Data	Methods	Features
Catlett & Al. [10]	Crime Hotspot Detection and Crime Forecasting	Crimes events in Chicago (2001-2016) and New York City (2006-2016)	DBScan for Crime Hotspot detection, ARIMA models for forecasting.	It is able to predict the expected crime number on a long-term time horizon for the detected crime-dense regions. It can detect crime-dense regions having any shapes. The exploited DBScan algorithm can detect wide crime-dense regions, or a large number of outliers.
Wei & Al. [29]	Crimes prediction distinguishing crimes categories	Crime events in Xiaogan and New York City, plus related demographic, POI, Road Network, and Land Use data.	Tensor representation, optimization through CANDECOMP/ PARAFAC decomposition, agglomeration nesting.	Differentiates crime categories, considers features other than crime data. Pre-defined set and shape of regions, custom-defined performance indexes.
Zubair & Al. [45]	Crime Hotspot Detection and Crime Classification Using Spatial Graph-Based Learning	Crimes events in Chicago (2001-2024)	Spatial Deep Graph Convolutional Network (GCN)	Detection of spatial dependencies by using graph-based neighborhood information, enabling accurate multi-class crime classification. It generates heat maps of high-risk areas without relying on temporal data.
Ahmad & Al. [37]	Crime hotspot detection	Crimes events in Vancouver (2006-2021), Crimes events in India (2018), ICT Police Crime Data (2009-2021)	CHART framework using Random Forest, Adaptive Synthetic Sampling (ADASYN), Kernel Density Estimation (KDE)	Integration of domain-specific features and KDE for accurate spatial hotspot visualization. Using Random Forest and ADASYN
Cesario & Al. [14]	Spatio-temporal crime forecasting in multi-density hotspots	Crimes events in Chicago (2001-2019)	CHD for Crime Hotspot detection, SARIMA and LSTM models for crime forecasting.	Detection of multi-density and multi-shape crime hotspots, specific crime forecasting model for each hotspot. Correlation with other urban events not considered.
Han & Al. [22]	Crimes prediction with feature extraction	Crimes events in Chicago (2014-2020)	Long Short-Term Memory (LSTM) and Spatial-Temporal Graph Convolutional Network (ST-GCN)	Detection of transition tendencies of crimes between neighboring communities over time. Communities with low number of crimes not considered.
Li & Al. [28]	Spatial-temporal prediction from sparse crime data	Crimes events in Chicago (2016-2017) and New York City (2014-2015)	Graph Neural Networks	Crime forecasts differentiated by categories. Predefined grid-cell hotspots could not detect dynamic crime distributions
Roshankar & Al. [39]	Spatio-temporal crime type prediction	Crimes events in Chicago (2012-2017)	STGNN (Spatio-Temporal Graph Neural Network), Random Forest, SVM, LSTM, ARIMA	Integration of spatial and temporal dependencies through a graph-based architecture, enabling accurate crime-type predictions. STGNN captures complex crime dynamics across time and space.

analysis, hotspot detection, and crime forecasting. Table 1 reports a detailed and critical comparison among these recent approaches and solutions proposed in the literature. The comparison takes into account four characteristics (goals, data, methods and features), as detailed in the following.

Goal of the Approach: The reported works are distinct with respect to the specific goals they want to achieve. Some approaches focus on crime prediction only [22], [29], others on hotspot detection [37], [45], and others offer comprehensive analysis workflows spanning from region discretization to event forecasting methods [10], [14], [28], [39].

Data: This dimension considers the type of data used to evaluate the various approaches. All methods have been tested on real-world crime datasets, mainly sourced from locations such as India [37], Chicago [10], [14], [22], [28], [39], [45], Vancouver [37], Xiaogan [29] and New York City [10], [28], [29]. In some cases, these datasets are enriched with additional contextual information [29].

Methods: The papers are distinguished by the different approaches based on the analytical and algorithmic strategies employed. For instance, the works in [10], [14] adopt various density-based clustering algorithms. The paper in [37] is mainly focused on crime hotspot discovery, and exploits Random Forest, ADASTN, and Kernel Density Estimation,

while [45] exploits Graph Convolutional Networks. The work in [29] exploits tensor representation and decomposition approaches. The methods in [39] utilize graph-based models to capture spatio-temporal correlations. Finally, other approaches rely on human-defined spatial boundaries and artificial neural networks to carry out crime prediction tasks [22], [28].

Main Features: Beyond the characteristics mentioned above, Table 1 also includes several distinctive features associated with the reviewed approaches. The methods discussed in [10], [14], [37], [45] stand out for their capability to autonomously identify hotspots of arbitrary shapes (be they circular, rectangular, or irregular). Moreover, the strategy presented in [10], [14] supports rolling-horizon forecasts, enhancing temporal predictions. Certain models are also designed to differentiate criminal predictions by category [28], [29], [39], [45], which may provide valuable information to law enforcement efforts. However, other approaches [22], [28], [29] are constrained to fixed or predefined spatial units (e.g., activity areas or grid cells), potentially limiting their adaptability to dynamic spatial crime patterns. Notably, although the method in [10] detects hotspots with diverse shapes, it also generates a considerable number of noise points. Finally, the techniques proposed in [29] benefit from integrating multiple data sources (e.g., crime statistics, demographic profiles, land use), which can provide insightful correlations between urban dynamics and criminal behavior. Nonetheless, this reliance on multi-source data could pose limitations when such data are not consistently available across all study areas.

With respect to the overall reviewed works, the present paper contribution distinguishes itself as it aims to comprehensively analyze the impact of different partitioning strategies, such as district-based or multi-density clustering-based, on the forecasting performance of crime events, by comprehensively analyze the results given by experiments in two real-world datasets, related to Chicago and Los Angeles crime events. We focus on density-based clustering algorithms since they can detect clusters having arbitrary shapes, giving also a view on the approaches (CHD, HDBSCAN) capable of detecting clusters having different densities. With respect to papers [6], [10], [14], that show forecasting performance of only a subset of the retrieved clusters, here we provide an all-around analysis on all the clusters, given the best parameter setting for each algorithm, found by performing parameter sweeping exploiting a set of several internal validation measures. The comparison is given by exploiting the same prediction horizon and the same aggregation time windows (one week).

III. PROBLEM DEFINITION AND GOALS

In this section, we set up a proper notation that will be used consistently throughout the paper and we define the goals of this research study. Let $T = \langle t_1, t_2, \dots, t_H \rangle$ be an ordered timestamp list, such that $t_h < t_{h+1}, \forall 0 \leq h < H$, and where all t_h are at equal time intervals (e.g., every hour,

day, week). Let $\mathcal{CD}$ be a crime dataset collecting crime events, $\mathcal{CD} = \langle cd_1, cd_2, \dots, cd_N \rangle$, where each cd_i is a crime record (or instance) represented by the triple $\langle latitude_i, longitude_i, t_i \rangle$, where $latitude_i$ and $longitude_i$ are the coordinates of the place, and t_i is the time (with $t_i \in T$) the event occurs at. Now, let us consider a future temporal horizon, $S = \langle t_s, t_{s+1}, \dots \rangle$, with $s > H$.

The aim of the analysis is to divide the entire city territory into a set of crime-based partitions, which may vary in size, crime count, and crime density, and to extract predictive models that accurately forecast the number of crimes within each partition, at a given timestamp $t_s \in S$. In particular:

- 1) discover a set $\mathcal{CP}$ of *crime partitions*, $\mathcal{CP} = \{CP_1, \dots, CP_K\}$, where a *crime partition* CP_k is a spatial area tailored to reflect its specific patterns of criminal behavior;
- 2) for each crime partition CP_i , compute the triple $\langle n_i, a_i, \sigma_i \rangle$, representing respectively the number of crimes, the area and the density of crimes characterizing the CP_i partition;
- 3) extract a set $\mathcal{F}_{crimes}(t)$ of *crime predictors*, $\mathcal{F}_{crimes}(t) = \{\mathcal{F}_{crimes}^1(t), \dots, \mathcal{F}_{crimes}^K(t)\}$, where each function $\mathcal{F}_{crimes}^k(t) : S \rightarrow \mathcal{R}$, given a timestamp $t_s \in S$ states the number of crimes $N \in \mathcal{R}$ that are predicted to happen in the *crime partition* $CP_k \in \mathcal{CP}$ at the timestamp t_s .

Given these definitions, the process involves two main tasks: identifying *crime partitions* and developing *crime predictive models*. The first task can be approached using various techniques (i.e., police district partitioning, machine-learning-based methods, etc.), while the second task can be addressed by the development of time series forecasting models.

IV. PARTITIONING AND FORECASTING APPROACHES

In this section, we present and summarize the different partitioning approaches that are the objective of the present investigation, as well as the predictive model exploited for crime forecasting.

A. APPROACHES FOR URBAN DATA PARTITIONING

This work compares the results of two kinds of partitioning strategies, that are *Human-Defined Partitioning* and *data-driven partitioning*.

Human defined partitions of a region are those provided arbitrarily by a human entity. This category includes grid cell partitioning, where a region is divided by a grid in different cells, each having the same width and length, and administrative boundaries, that is, dividing a region by means of boundaries defined by, e.g., zip codes or districts. Human defined partitioning are static, the partitions show regular shapes, and are usually defined by humans aiming at a particular goal, such as census data collection, electoral districts, resource allocation, and policy planning [30], [41].

Data-driven partitioning is instead provided by algorithmically analyzing data or events of interest, in general,

by exploiting clustering algorithms. This kind of strategy is more dynamic, partition boundaries intrinsically suited to the available data, and can show any kind of shape [5], [14].

The following provides a deeper view of human-defined partitioning and summarizes two data-driven partitioning algorithms: DBSCAN, HDBSCAN, and CHD.

1) HUMAN DEFINED PARTITIONING

Human-defined partitions of a region are those provided arbitrarily by a human entity.

In urban events prediction, the main exploited human-defined partitioning methods can be categorized in:

- *Geometric partitioning*. This involves partitioning regions using a geometric and regular pattern, that is, subdividing regions in a grid of regular squared cells, with a pre-fixed size. This approach produces partitions having the same extension, but a very variable number of events.
- *General administrative boundaries*. This involves partitioning a region by exploiting administrative boundaries that do not consider the specific kind of urban events under analysis. This can include partitioning events by zipcodes or city districts.
- *Specific administrative boundaries*. This involves partitioning data by exploiting pre-existing administrative boundaries that were designed and tailored for a purpose that is closely related to the urban events under analysis. This is the case of, e.g., partitioning data by police districts for crime events analysis, and by land usage regulation for economic data analysis.

Within the scope of the present paper, we compare a specific administrative boundaries partitioning strategy against two data-driven partitioning strategies.

2) HDBSCAN

The HDBSCAN algorithm [8] computes a complete clustering hierarchy composed of all possible density-based clusters for a large range of density thresholds. Then, it chooses the clustering model that maximizes the overall stability of the extracted clusters. To build such a hierarchy, the HDBSCAN starts from the concept of *mutual reachability distance* between two core points p and q and a given value for a parameter min_pts . The *mutual reachability distance* is defined as the minimum ϵ radius such that p and q are mutually density-reachable. Given two data points p and q , they are mutually density reachable if there exists a chain of points, starting from q and ending at p , such that each point in the chain has a density greater than or equal to a specified minimum density threshold, min_pts , and also exists a similar chain that starts from p and ends in q .

HDBSCAN works as follows. First, it builds the clustering hierarchy by computing a *mutual reachability graph*, which is a complete graph where each vertex is a data point, and each edge is weighted with the mutual reachability distance of the linked couple of points. Then, a minimum

spanning tree is computed on that graph, integrated by adding (for each vertex) a self-edge, weighted by the *core-distance* of the related data point. The tree is processed by removing the edges in decreasing order w.r.t. their weight. For each removal, the two involved edges are labeled as roots of a new pair of clusters, or noise if the generated component has not any edge. A variation of the summarized algorithm considers also a given minimum cluster size (*min_cluster_size* parameter), which avoids the generation of clusters having a size lower than *min_cluster_size*. Given the clustering hierarchy, a clustering is extracted which maximizes the overall stability. The notion of stability is derived from the notion of *excess of mass* [36]. HDBSCAN is able to compute the clustering hierarchy and extract the clusters in $O(n^2)$ time. In this work, we exploited the implementation provided by the scikit-learn library.¹

3) CITY HOTSPOT DETECTOR

The *City Hotspot Detector* (CHD) algorithm [16] is a multi-density based clustering algorithm that has been purposely designed for processing urban spatial data. The algorithm is composed of several steps, as follows. First, given a fixed *min_pts*, the reachability distance for each point is computed and exploited as an estimator of the density of each data point. Then, the points are sorted w.r.t. their estimated density and the density variation between each consecutive couple of points in the ordered list is computed. The obtained density variation list is then smoothed by applying a rolling mean operator considering windows of size s . By considering the such smoothed density variation, the points are then partitioned into several *density level sets*. For each density level set, a different ϵ value is estimated. Finally, each set is analyzed by the DBSCAN algorithm. Specifically, each instance takes in input a specific ϵ value computed for the analyzed density level set. The set of clusters detected for each partition constitutes the final result of the CHD algorithm. The CHD algorithm runs with $O(n \cdot \log(n))$ time complexity, where n is the size of the processed dataset. A cluster analysis with the CHD algorithm requires the tuning of more parameters (three) w.r.t. HDBSCAN. In this work, we exploited the CHD implementation that was made publicly available from the authors on git repository.²

4) DBSCAN

The DBSCAN (density-based spatial clustering of applications with noise) [19] algorithm is the foundational algorithm in the family of density-based clustering techniques. It was specifically designed to handle large datasets that contain noise and can identify clusters of arbitrary shape without requiring prior knowledge of the number of clusters. Shortly, DBSCAN leverages the concepts of *core points*,

¹The scikit-learn HDBSCAN documentation: <https://scikit-learn.org/stable/modules/generated/sklearn.cluster.HDBSCAN.html>

²Git repository for CHD implementation: <https://gitlab.com/chd3/parallel-chd>

density-reachability, and *density-connectivity*. The algorithm requires two parameters: ϵ , which defines the neighborhood radius, and *minPts*, the minimum number of points required to form a dense region. A point is a *core point* if at least *minPts* points lie within its ϵ -neighborhood (neighborhood of radius ϵ). A point p is directly density-reachable from another point q if (i) q is a core point and (ii) p lies within the ϵ -neighborhood of q . Two points p and q are *density-reachable* if there exists a chain of directly density-reachable points connecting q and p . Moreover, two points p and q are said to be *density-connected* if (i) there exists a core point o from which both points (p and q) are density-reachable. Clusters are formed by iteratively linking density-connected points, and all points reachable in this way from a cluster member are included in that cluster. Points that cannot be associated with any cluster are classified as noise. With an appropriate spatial indexing method for the ϵ -neighborhood searches, DBSCAN can process a dataset of size n in $O(n \cdot \log(n))$ time. In this work, we exploited the implementation provided by the scikit-learn library.³

B. PREDICTION WITH SARIMA

Multiple regression models have been defined with the goal of forecasting a variable of interest using a linear combination of predictors [24]. In particular, in an *auto-regression model*, the variable of interest is forecasted using a linear combination of its past values (the term auto-regression indicates that it is a regression of the variable against itself), while a *moving average model* uses past forecast errors in a regression-like model. Sometimes, as a preliminary step to the regressive analysis, time series need a *differencing* transformation to stabilize the mean of a time series and so eliminating (or reducing) trend and seasonality. A combination of differencing, auto-regression and moving average methods is known as *AutoRegressive Integrated Moving Average* models [24]. Such regression model is referred as *ARIMA*(p, d, q), where the order of the model is stated by three parameters: p (order of the auto-regressive part), d (degree of first differencing involved) and q (order of the moving average part).

In order to deal with seasonality, the classical ARIMA processes have been generalized and extended by the SARIMA (i.e., *Seasonal ARIMA*) models. A SARIMA model is formed by including additional seasonal terms (modeling a seasonal component that repeats with a given periodicity) in the classic ARIMA models previously introduced. The seasonal part of the model consists of terms that are very similar to the non-seasonal components of the model. The additional seasonal terms are simply multiplied by the non-seasonal terms. A seasonal ARIMA model is referred as *SARIMA*(p, d, q)(P, D, Q) $_m$, where m is a periodicity

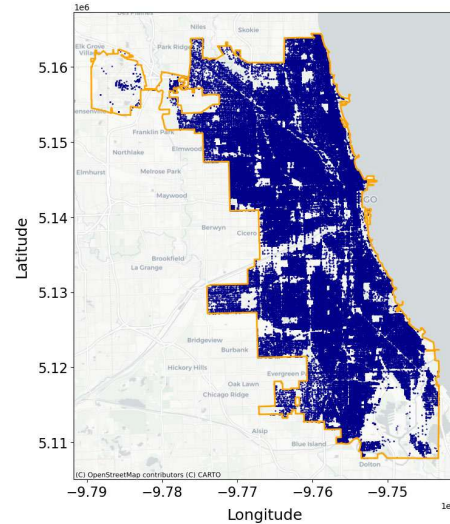


FIGURE 2. Geo-localized crime events in Chicago (2001–2019).

factor. The implementation exploited in this work is the one provided by the pmdarima package.⁴

V. CHICAGO CRIME DATA: EXPERIMENTAL RESULTS

To evaluate the performance of the three approaches outlined in Section IV-A, namely *Police-Districts*, *HDBSCAN*, *CHD* and *DBSCAN*, we conducted a comprehensive experimental analysis using the full territory of Chicago as our real-world case study. This analysis compares the effectiveness of the territorial partitions formed by police districts and crime hotspots, assessing each approach's partition quality and crime forecasting accuracy within each district/hotspot. The rest of this section is organized as follows. Section V-A describes the area selected for the analysis and the gathered data, Section V-B describes how we have estimated the optimal parameter setting for the clustering algorithms, Section V-C reports some details (area, shapes, number of crimes, etc.) about the pre-defined *police-districts* and the *crime hotspots* detected by the three different approaches, and Section V-D describes the evaluation of the forecasting accuracy achieved by SARIMA models across the partitions created by the three different partitioning techniques.

A. CHICAGO: DATA DESCRIPTION

The data used for our experimental evaluation was gathered from the *Chicago Data Portal*, an open-access data exploration and search platform developed and managed by the City of Chicago.⁵ Specifically, we utilized crime data from the 'Crimes - 2001 to Present' dataset, a real-world collection detailing criminal events in Chicago from 2001 to present. Each entry in this dataset represents a geo-located crime event, identified by several attributes including time,

³The scikit-learn DBSCAN documentation: <https://scikit-learn.org/stable/modules/generated/sklearn.cluster.DBSCAN.html>

⁴The pmdarima home page: <https://alkaline-ml.com/pmdarima/modules/classes.html>

⁵Chicago Data Portal: <https://data.cityofchicago.org/>

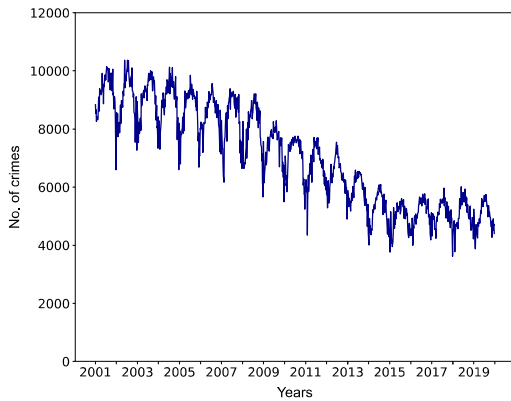


FIGURE 3. Number of crimes vs time (2001–2019).

longitude-latitude coordinates, police district, etc.⁶ In this study, we perform our experiments on the entire Chicago area, with its boundaries and geo-localized crime events illustrated in Figure 2. The city's perimeter measures approximately 96 KM, encompassing an area of around 589'9 KM². From the 'Crimes - 2001 to Present' dataset, we collected all crime records that occurred in Chicago over a span of 20 years, from January 2001 to December 2019. This collection includes 6,979,257 crime events, averaging 7,043 incidents per week. The total size of the whole dataset is 550 MB. Chicago's diverse neighborhoods exhibit varying population densities and commercial activity, thus in turn leading to different crime density patterns across areas. This diversity makes the city an especially compelling case for spatio-temporal crime forecasting, as it allows us to examine crime trends and model predictive patterns across neighborhoods with differing crime dynamics.

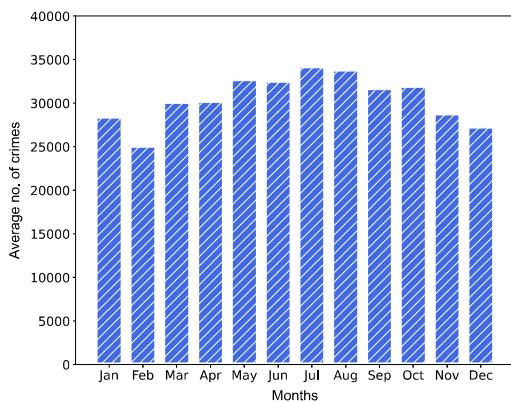


FIGURE 4. Distribution of the number of crimes by month.

Figures 3 and 4 present an initial descriptive analysis of the collected crime data, revealing key trends and patterns. Figure 3 plots the number of collected crimes versus the

⁶Crimes from 2001 to Present: <https://data.cityofchicago.org/Public-Safety/Crimes-2001-to-present-Dashboard/5cd6-ry5g>

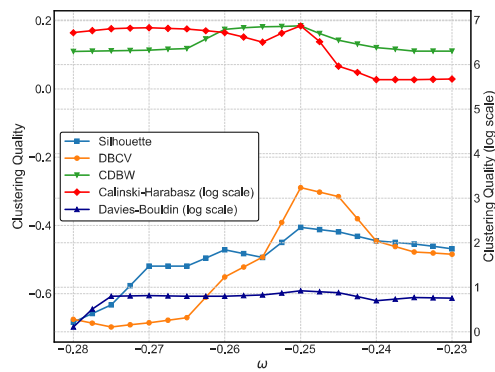
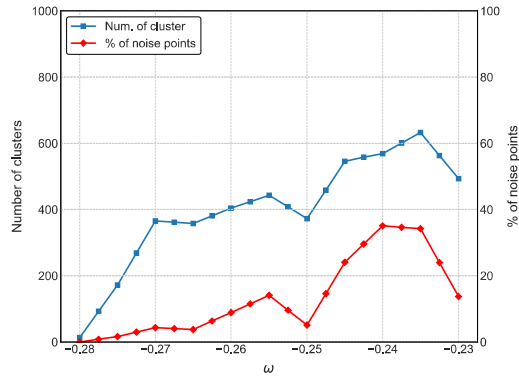
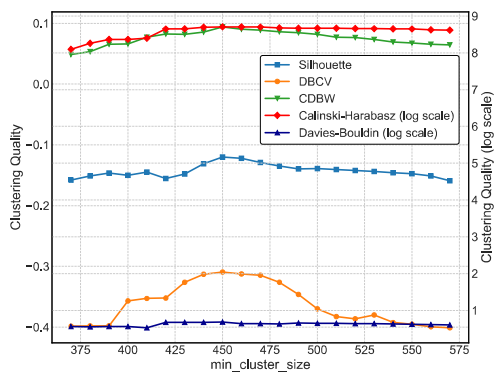
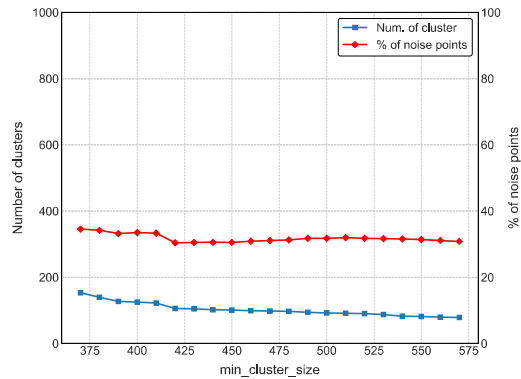
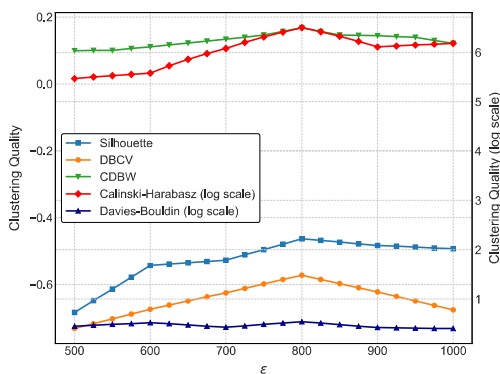
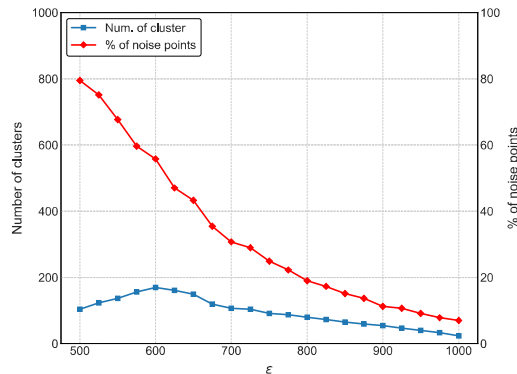
time of observation. The plot immediately reveals some interesting insights. First, there is an evident *decreasing trend* in crime numbers from 2001 through 2015, after which the rate stabilizes from 2015 to 2019. Additionally, the data reveals a distinct *seasonal pattern* within each year, with seasonal crime peaks diminishing as the overall crime trend decreases. The plot further shows that crime rates typically increase in late Spring, peak during Summer, decline through Autumn, and reach their lowest numbers in Winter. Figure 4 shows the distribution of the average number of crimes by month, thus providing a clearer picture of these seasonal trends. This histogram highlights significant fluctuations across months, with the highest number of crimes occurring in July (averaging 34,237 crimes) and the lowest one in February (averaging 25,126 crimes). Together, these visualizations underscore significant temporal variations and seasonal dynamics within Chicago's crime data.

Before performing the specific data analysis task, a pre-processing phase was carried out, in which only the columns relevant to the analysis have been filtered, namely latitude, longitude, and timestamp. In order to ensure geographical consistency of the study area, crimes that did not fall within the perimeter and administrative area of Chicago were removed from the dataset (less than 0.5% of the dataset). These values included coordinates equal to zero, probably marked with this value when the precise location of the crime was unknown or unavailable (missing values). After the filtering phase, we converted the latitude and longitude variables into Cartesian coordinates (i.e., x and y), which ensures the correct use of Euclidean distances exploited during the clustering process.

B. CHICAGO: ESTIMATING OPTIMAL PARAMETERS FOR HDBSCAN, CHD AND DBSCAN

As described above, in this research study crime partitions are identified using three distinct approaches: Police-Districts, HDBSCAN, CHD, and DBSCAN. While the Police-Districts method relies on a pre-defined territorial division established by law enforcement, the other three approaches leverage density-based clustering algorithms, providing a data-driven perspective tailored to the spatial distribution of crime events.

To conduct a fair comparative analysis of the four approaches, it is essential to fine-tune the input parameters of the clustering algorithms (i.e., HDBSCAN, CHD, and DBSCAN) to ensure the detection of high-quality city hotspots in terms of separability, compactness, and significance. Specifically, the CHD algorithm requires the setting of k , ω and s parameters, HDBSCAN requires the setting of $\minPts$ and $\min_cluster_size$, and DBSCAN requires the setting of $\minPts$ and ϵ parameters. To identify the optimal configurations and show the best results achievable by the algorithms, we adopted a parameter-sweeping methodology, that is, we run several instances of the algorithms by varying their input parameters. Then, we select the best result

(a) Clustering quality vs ω .(b) Number of hotspots and noise points vs ω .**FIGURE 5.** Chicago crime data. CHD: clustering quality, number of hotspots and noise points.(a) Clustering quality vs $min_cluster_size$.(b) Number of hotspots and noise points vs $min_cluster_size$.**FIGURE 6.** Chicago crime data. HDBSCAN: clustering quality, number of hotspots and noise points.(a) Clustering quality vs ϵ .(b) Number of hotspots and noise points vs ϵ .**FIGURE 7.** Chicago crime data. DBSCAN: clustering quality, number of hotspots and noise points.

(evaluated in terms of clustering quality) achieved by each algorithm, which best suits our application scenario and the considered dataset.

Since crime data are inherently unlabeled, clustering quality was assessed using internal validation measures [33], which evaluate the clustering structure without reference

to external labels. Internal indices assess cluster quality based on intra-cluster cohesion and inter-cluster separation. To do so, the following set of *internal indexes* are here employed: Silhouette [40], DBCV [35], CDBW [20], Calinsky-Harabaz [7], Davies-Bouldin [17]. These indices are widely used to measure clustering quality in terms of compactness, separation, density, and the number of clusters when no external labels are available [33].

The first set of experimental results is presented in Figure 5, which shows the performance of the CHD algorithm as the parameter ω varies from -0.28 to -0.23 . Specifically, Figure 5a highlights how the above mentioned internal indexes, used to assess clustering quality, change with different values of ω . These indexes provide insight into the compactness, separability, and overall structure of the detected clusters. The results indicate that the clustering quality is highly sensitive to the choice of ω , with the optimal value clearly estimated as $\omega^* = -0.25$ in our tests. On the other side, Figure 5b shows the variation in the number of noise points (red curve) and the number of detected hotspots (blue curve) as ω changes. Noise points are data instances that do not meet clustering the criteria for falling into any of the detected clusters, and are thus treated as outliers by the algorithm. The number of detected hotspots reflects the algorithm's ability to balance clustering separability and compactness effectively. Notably, at $\omega^* = -0.25$, CHD detects 354,546 noise points (equal to 5.08% of all points) and identifies 373 clusters.

The experimental results for HDBSCAN, the second density-based clustering algorithm employed in this study, are presented in Figure 6. This figure evaluates the clustering quality as the parameter *min_cluster_size* varies from 375 to 575. Specifically, Figure 6a illustrates how clustering quality changes with different values of *min_cluster_size*. The results identify the optimal parameter value as *min_cluster_size*^{*} = 450, which achieves the highest clustering quality according to the internal validation measures. On the right side, Figure 6b shows that both the number of clusters and noise points have a slightly decreasing trend when *min_cluster_size* increases, indicating that larger cluster size thresholds lead to fewer but more compact clusters. At the optimal value *min_cluster_size*^{*} = 450, HDBSCAN identifies 2,132,861 noise points (equal to 30.56% of all points) and detects 101 clusters.

The results regarding the DBSCAN algorithm are reported in Figure 7, considering the ϵ parameter ranging between 500 to 1000. By observing Figure 7a, we can highlight that the best ϵ parameter setting in this case is $\epsilon = 800$, that achieves the best internal clustering quality indexes. This corresponds to identifying 1,326,757 noise points (equal to 19.1% of all points) and detecting 80 clusters.

As reported above, we conducted extensive experimental tests to identify the parameter settings that produce the highest-quality clusters (i.e., crime hotspots). For such a reason, in the following we present the results achieved by CHD using the parameter settings $\omega = -0.25$, $k = 64$, and

$s = 5000$, by HDBSCAN with *min_cluster_size* = 450 and *min_points* = 16, and by DBSCAN with *minPts* = 16 and $\epsilon = 800$. These parameter configurations were selected as they best align with the specific requirements of our application scenario and the characteristics of the dataset, as demonstrated in the results shown above.

C. CHICAGO: CRIME HOTSPOTS DETECTED BY POLICE-DISTRICTS, HDBSCAN, CHD AND DBSCAN

Now, we analyze in more detail the crime hotspots identified in the city of Chicago. Figure 8 presents the partitions identified by the four approaches: Police-Districts, HDBSCAN, CHD, and DBSCAN. Each partition is depicted with distinct colors, while noise points, representing outliers, are marked in black. Specifically, Figure 8a illustrates the predefined territorial divisions into police districts. These divisions are established by law enforcement and publicly accessible through the *Chicago Data Portal*. Further information about district partitioning is available at https://data.cityofchicago.org/Public-Safety/Crimes-2001-to-Present/ijzp-q8t2/about_data, and the corresponding district boundaries can be found at <https://data.cityofchicago.org/Public-Safety/Boundaries-Police-Districts-current-/fthy-xz3r>. The district shapes in Figure 8a are drawn as regular polygons, delineated based on major streets and neighborhood borders. In contrast, Figures 8c, 8b, and 8d show the clusters identified by the CHD, HDBSCAN, and DBSCAN algorithms. These clustering methods identify dense regions of data points, enabling the detection of clusters with arbitrary shapes while also recognizing noise points as outliers in the dataset. Table 2 summarizes the key characteristics of each partitioning approach, including the input parameters and the number of detected hotspots (for HDBSCAN, CHD, and DBSCAN), the number of districts (for Police-Districts), and the percentage of noise points. By analyzing the results presented in Figure 8 and Table 2, two main observations emerge:

(i) **CHD detects an higher number of crime partitions compared to HDBSCAN, DBSCAN, and Police-Districts.** As shown in Table 2, Police-Districts divides the entire territory into 22 predefined districts, while DBSCAN, HDBSCAN, and CHD identify 80, 101, and 373 hotspots, respectively. DBSCAN, HDBSCAN and CHD offer finer-grained partitions on average, detecting smaller and more localized crime-prone areas compared to the broader police district boundaries, even if they also detect wider areas with outlier points or low crime-density rates. The provided granular segmentation provides a more precise identification of neighborhoods affected by crimes, potentially enhancing crime analysis and intervention strategies. Some further considerations can be done in terms of noise points, i.e., data points that can not be assigned by DBSCAN, HDBSCAN and CHD to a specific hotspot because density requirements are not satisfied (colored in black in Figures 8d, 8b and 8c). According to Table 2, CHD classifies only about 5.08% of data instances as noise,

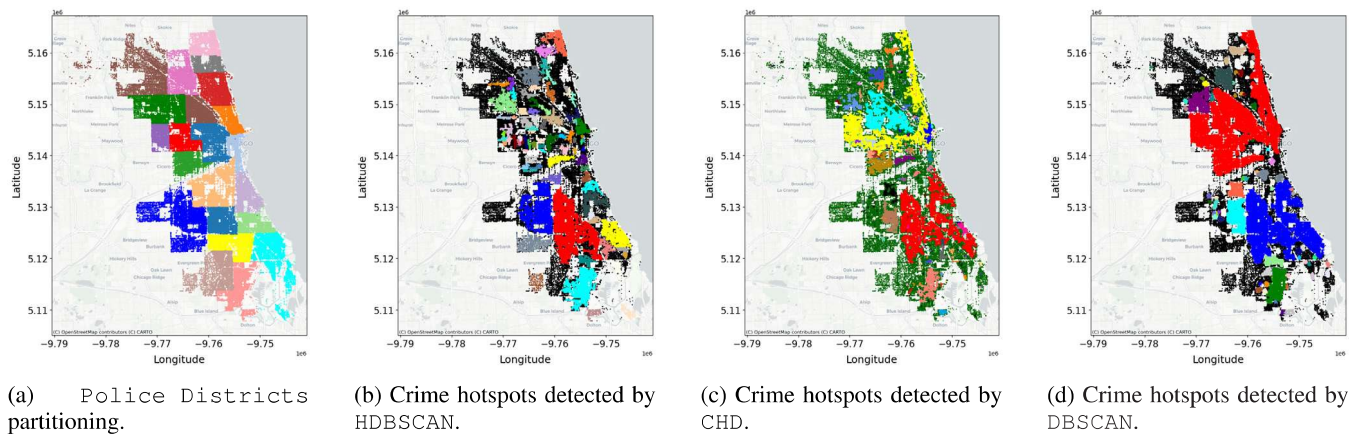


FIGURE 8. Crime partitions detected in Chicago.

a substantial improvement over HDBSCAN, which marks 30.56% of data instances as noise, and DBSCAN that marks 19.1% as noise. This difference highlights the effectiveness of CHD in minimizing unclustered data and providing more comprehensive coverage of crime hotspots.

(ii) **Police-Districts, DBSCAN, HDBSCAN and CHD detect very different partitions of the city.** As illustrated in Figures 8a, 8d, 8b, and 8c, these techniques generate partitions that differ significantly in terms of area coverage, shape, boundaries, and the number of points assigned to each partition. Specifically, *Police-Districts* relies on predefined administrative boundaries, dividing the city into a few large and regular polygonal shapes. These boundaries are static and determined by major streets and neighborhood divisions rather than the actual spatial distribution of crime events. In contrast, DBSCAN, HDBSCAN and CHD utilize density-based clustering to dynamically identify partitions based on crime data. This data-driven approach leads to partitions that reflect the spatial patterns of crime, resulting in more varied hotspot shapes and areas. For instance, CHD identifies a large crime hotspot in the south-central area of the city (highlighted in red in Figure 8c), which spans across multiple police districts as defined by *Police-Districts* and is also only partially detected by HDBSCAN. Similarly, in the central-northern part of the city, CHD detects several distinct crime hotspots (colored in magenta, green, light blue, and yellow in Figure 8c) that partially overlap multiple police districts shown in Figure 8a. Moreover, it can be observed that DBSCAN tends to produce broader clusters that encompass those identified by the other two clustering methods, e.g., the red northern cluster detected by DBSCAN is split into multiple, more refined clusters by both CHD and HDBSCAN. Interestingly, there are also examples of convergence among the approaches. For example, the southernmost police district (colored in pink in Figure 8a) corresponds approximately to crime hotspots identified by both HDBSCAN (cyan-colored in Figure 8b) and CHD (pink-colored in Figure 8c). Such overlaps suggest that while the methods diverge in their

general partitioning strategies, certain regions of the city exhibit spatial crime patterns consistent across different approaches.

TABLE 2. Overview of the partitioning metrics obtained by the different techniques.

Partitioning technique	Input parameters	# hotspots/districts	# noise points
District	—	22	—
HDBSCAN	$min_cluster_size = 450$, $min_points = 16$	101	30.56%
CHD	$\omega = -0.25$, $k = 64$, $s = 5000$	373	5.08%
DBSCAN	$\varepsilon = 800$, $min_points = 16$	80	19.01%

To have a complete overview of the results achieved by the four partitioning approaches, we analyze here the characteristics of the district/hotspots detected, in terms of number of crimes, area (KM²) and crime density (number of crimes/KM²). In Figure 9 we show, for each partitioning strategy, the cumulative distribution of the number of crimes in each detected partition. As an instance, we can say from Figure 9a that the 30% of the districts contains at most 300K crime events.

From the figures, we can highlight that considering the *Police-Districts* partitioning, the number of crimes collected in each pre-given district ranges from 200K to around 500K, and the occurrence rate of such values is almost balanced among the districts, since the cumulative distribution grows with a linear trend starting at 200K crimes. On the other side, the majority of hotspots (more than 90%) detected by DBSCAN. HDBSCAN and CHD contain less than 200K, 50K and 100K crimes, respectively. Similar considerations can be made considering areas (Figure 10) and crime density (Figure 11) of the districts/hotspots. For example, Figure 10 shows that police districts have areas distributed between 10 KM² and 90 KM², with around the 90% of districts smaller than 40 KM²; the majority of hotspots detected by HDBSCAN and DBSCAN have extensions lower than 20 KM²; finally, the most of hotspots detected

by CHD have an area lower than 5 KM². Furthermore, from Figure 11, we can show how DBSCAN considers only partitions having an average density spanning from 8,000 to 16,000 crimes/KM², while CHD and HDBSCAN can detect clusters characterized by a wider range of densities. This is due to the fact that, differently from DBSCAN, the latter approaches can detect clusters characterized by very different densities. The Policy-District partitioning strategy here shows an almost uniform distribution between 2,500 and 30,000 crimes/KM² of the clusters' density.

D. CHICAGO: CRIME FORECASTING RESULTS

In this section, we discuss how the different partitioning techniques impact crime forecasting performance.

To perform such an analysis, we compare the results by exploiting two well-known metrics for forecasting tasks: Mean Absolute Error (MAE) and Mean Absolute Percentage Error (MAPE). The MAE index expresses the mean absolute error of the forecasted values w.r.t. the actual values. The MAE is defined as:

$$MAE = \frac{1}{n} \sum_{t=1}^n |A_t - F_t| \quad (1)$$

where A_t is the actual value at time t and F_t is the forecasted value at time t .

In our case, as also suggested in [5], since different partitions can comprehend a different number of events, we also use the MAPE index, which is a relative metric that weights the point-to-point error w.r.t. actual values. The MAPE is defined as:

$$MAPE = \frac{1}{n} \sum_{t=1}^n \left| \frac{A_t - F_t}{A_t} \right| \quad (2)$$

For each cluster generated by each partitioning technique, we computed the time series expressing the number of crime events in the cluster, aggregated by weeks. To conduct a fair comparison, we fitted a different SARIMA regression model for each obtained time series, with a forecasting time step of one-week-ahead, using events between 2001 and 2016 as training set. Similarly to cluster model fitting, in this case we exploited a grid search technique to find the best parameter setting for SARIMA for each analyzed series, exploiting the MAPE as the performance metric to minimize. MAE and MAPE indexes are then computed for each time series, considering events between 2017 and 2019 as test.

To draw a sketch of the obtained results, Figure 12 shows the observed time series vs the forecasted ones for three random clusters, one for each partitioning method. Here we qualitatively observe that the fitted SARIMA models are capable of forecasting the observation trends. The quantitative results are instead shown in Figures 13 and 14, which report the MAPE and MAE metrics for each partitioning technique and for each detected cluster, ordered by area, number of crimes, and density. In other words, each depicted point represents a cluster defined in the space of the

SARIMA forecasting performance metric (MAPE or MAE) for each considered dimension (area or number of crimes or density). Based on the information provided, we can draw the following considerations.

Based on the information provided, we can draw the following considerations.

Forecasting performance depends on cluster size in terms of number of crimes. Overall, as expected, the forecasting quality depends on cluster size, that is, the more the data, the more the models perform. This is highlighted in Figure 13, by observing how the MAPE grows as the cluster size decreases, and such a phenomenon is particularly evident for CHD, HDBSCAN, and DBSCAN partitioning techniques. The district-based partitioning seems to not respect this trend, but this is due to the unavailability of clusters having less than one hundred thousand crime events. The highlighted consideration also seems to hold when analyzing the forecasting quality regarding the cluster area. This is because, in our experiments, cluster size and cluster area are linearly correlated.

Multi-density-based partitioning achieves higher performance than district-based partitioning. HDBSCAN-based and CHD-based partitioning approaches outperform district-based partitioning in terms of forecasting accuracy. This is suggested by the fact that, for all clusters having comparable sizes (greater than 100k events), both MAPE (Figure 13) and MAE (Figure 14) metrics are lower in HDBSCAN-based and CHD-based partitioning techniques w.r.t. to the District-based partitioning technique. When considering fewer clusters, that is, fewer than 100k events, we also observe that the decreasing trend of MAE is confirmed for CHD- and HDBSCAN-based partitions. The same consideration can not be drawn for the DBSCAN partitioning approach, since the results in terms of both MAPE and MAE are similar to the police-district strategy when considering clusters having comparable size. This behavior can be explained by the fact that multi-density partitioning methods can detect areas that are more homogeneous in terms of crime density with respect to district-based partitioning.

No direct relation between average cluster density and performance. Even if we observed that both multi-density-based partitioning methods perform better than the other approaches, it is worth noting that they do not show a clear relation between densities and MAPE, even if the district-based partitioning MAPE performance seems to decrease when the event density increases. The same trend is confirmed when analyzing MAE metrics, with some exceptions for CHD- and HDBSCAN-based partitioning, due to the presence of clusters that have a similar density (around 20k events per km²), but cover an increasing area. The better performances of multi-density based partitioning, previously observed in this section, are due to the fact that both CHD and HDBSCAN are capable of discovering partitions each having a more uniform crime density w.r.t. police districts.

CHD and HDBSCAN have different strengths. By comparing the results provided by CHD- and HDBSCAN-based

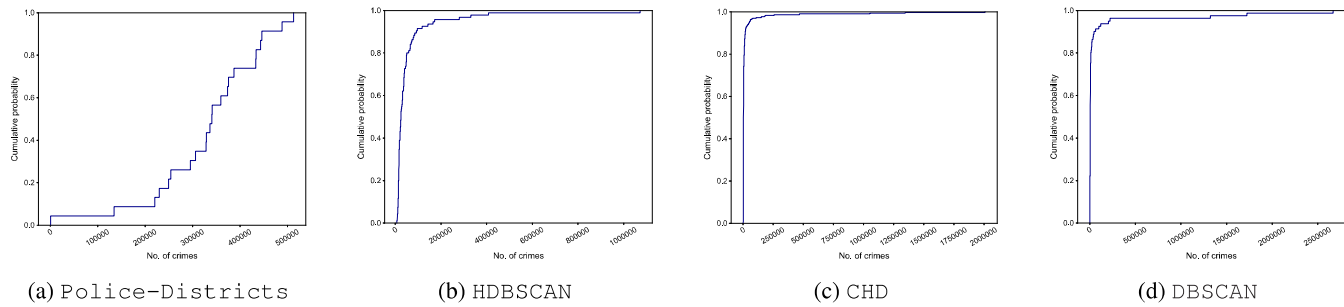


FIGURE 9. Cumulative distribution of the number of crimes in each district/hotspot, for the three approaches.

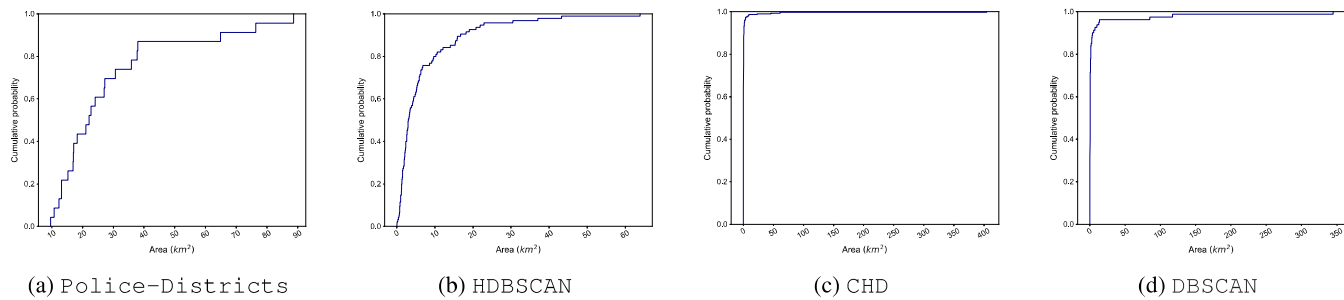


FIGURE 10. Cumulative distribution of the area in each district/hotspot, for the three approaches.

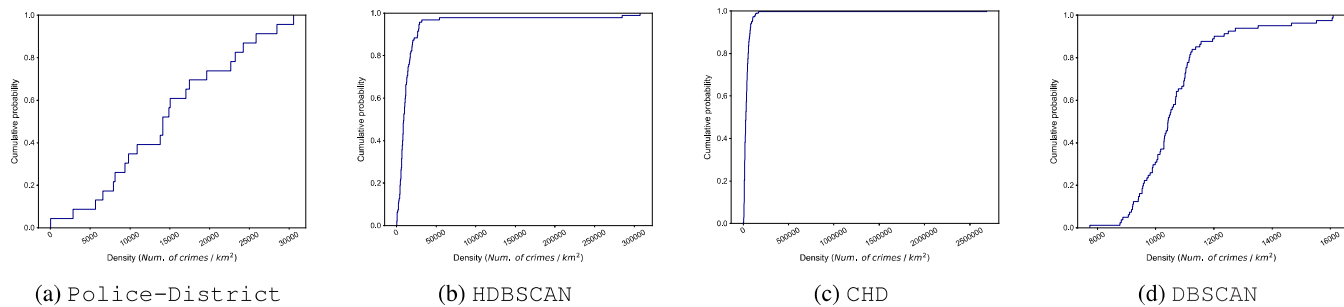


FIGURE 11. Cumulative distribution of the crime density in each district/hotspot, for the three approaches.

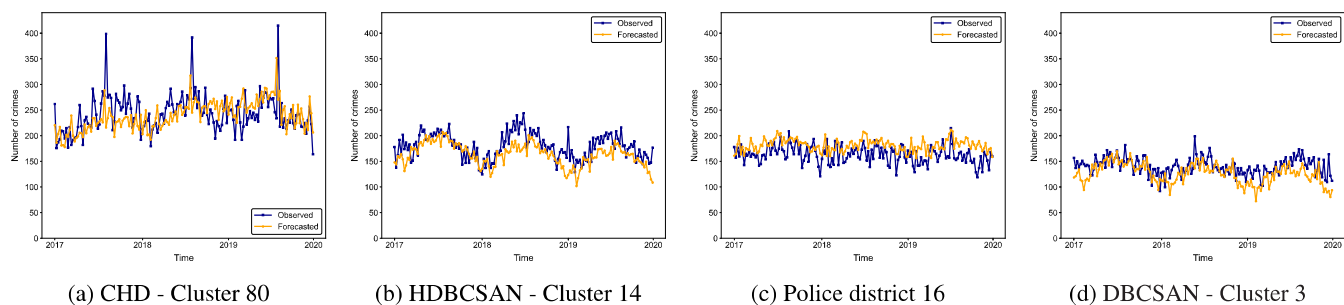


FIGURE 12. Observed vs Forecasted: compare the cluster with high number of crimes.

partitioning, it is possible to note that, when considering both cluster size and area, the results obtained by CHD are, in general, slightly better, showing lower MAPE and MAE values, and also that CHD-based partitioning results

seem to be more predictable, as they show less variable trends. This can be observed evaluating the MAPE trends in Figure 13, considering areas smaller than 25km^2 and sizes lower than 100k. However, considering the MAE (Figure 14),

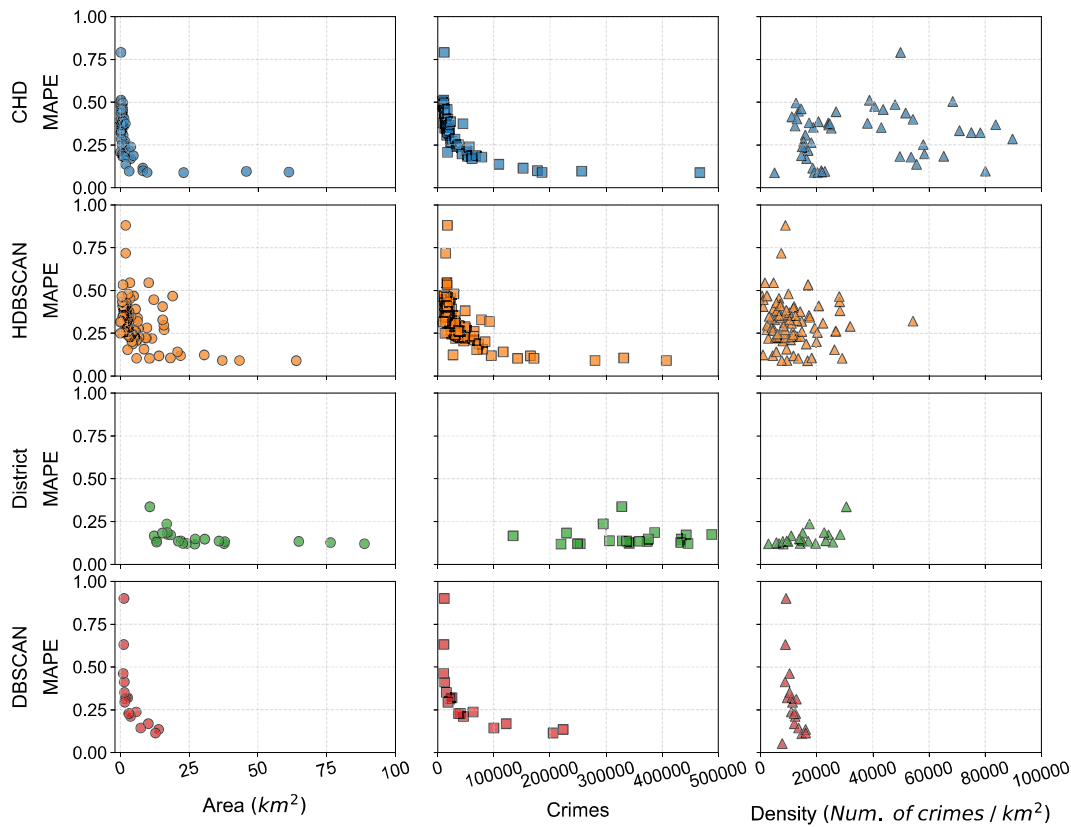


FIGURE 13. Scatterplot MAPE vs Cluster area, number of crimes, density.

we observe that HDBSCAN performs slightly better than CHD in clusters having a size greater than 100k.

As a final observation, we grouped the three most similar clusters of each partitioning approach, in terms of (i) number of crimes (Table 3), (ii) number of crimes and cluster area (Table 4), and (iii) number of crimes and average cluster density (Table 5).

Figure 15 shows the forecasting performances of each group of clusters exploiting the MAPE index. These results also confirm the better performance of multi-density-based partitioning techniques compared to the District-Based and the DBSCAN-based partitioning. Also, it is possible to confirm that CHD and HDBSCAN performances are similar.

VI. LOS ANGELES CRIME DATA: EXPERIMENTAL RESULTS

This section presents a comparative analysis of the four territorial partitioning approaches evaluated in this study, i.e., Police-Districts, HDBSCAN, CHD, and DBSCAN, performed on the entire territory of Los Angeles (exploited as second real-world case study). The goal of this analysis is to assess and compare the quality of the spatial partitions generated by each method and their effectiveness in supporting crime forecasting tasks. Specifically, the comparison focuses on the spatial coherence of the partitions and the forecasting accuracy achieved within each district or

hotspot. Following the structure adopted in the previous case study, this section is organized into three subsections: Section VI-A provides a description of the geographic area under investigation and the corresponding collected data, Section VI-B reports some details about parameter setting values, Section VI-C offers details on the official police districts and the crime hotspots identified by the clustering algorithms, and finally, Section VI-D evaluates and compares the forecasting performance across the different partitioning strategies.

A. LOS ANGELES: DATA DESCRIPTION

The dataset used in our experiments was obtained from the *Los Angeles Data Portal*, an open-access platform maintained by the City of Los Angeles.⁷ Specifically, we utilized the ‘Crimes - 2010 to 2019’ dataset, which contains geo-referenced records of criminal events occurred in Los Angeles from January 2010 to December 2019. Each crime is described by several attributes, including attributes such as timestamp, location coordinates, and police district. Our analysis focuses on the entire Los Angeles metropolitan area, as illustrated in Figure 16, covering approximately 1,214.9 KM^2 with a perimeter of about

⁷Crime Data from 2010 to 2019: https://data.lacity.org/Public-Safety/Crime-Data-from-2010-to-2019/63jg-8b9z/about_data

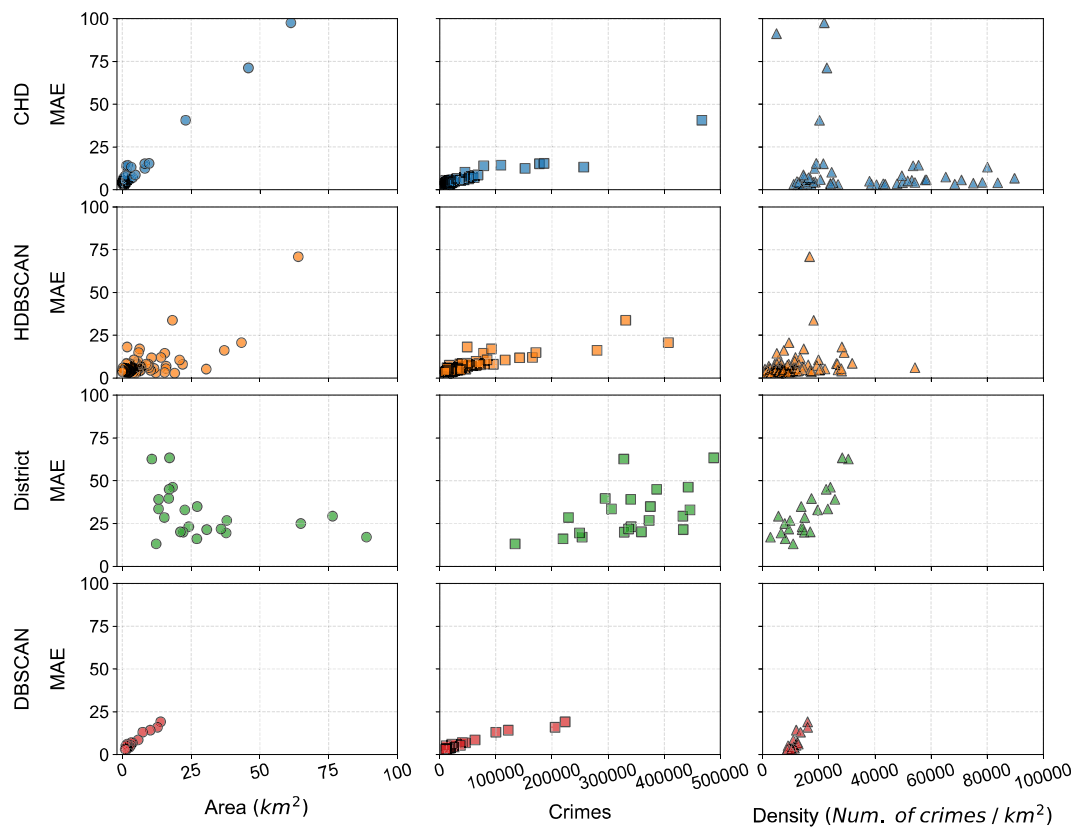


FIGURE 14. Scatterplot MAE vs Cluster area, number of crimes, density.

TABLE 3. The most similar clusters in terms of number of crimes.

CHD		HDBSCAN		DISTRICT		DBSCAN	
Cluster ID	Num. of crimes	Cluster ID	Num. of crimes	Partition ID	Num. of crimes	Cluster ID	Num. of crimes
80	256,469	14	280,241	16	253,646	3	223,452
282	177,701	62	171,400	17	219,697	4	205,588
283	152,051	18	142,443	20	134,264	7	122,190

TABLE 4. The most similar clusters in terms of number of crimes and area.

CHD			HDBSCAN			DISTRICT			DBSCAN		
Cluster ID	Num. of crimes	Area	Cluster ID	Num. of crimes	Area	Partition ID	Num. of crimes	Area	Cluster ID	Num. of crimes	Area
277	466,965	23.00	28	331,248	18.19	11	487,989	17.15	3	223,452	13.91
284	185,769	9.70	18	142,443	10.60	24	229,532	15.27	4	205,588	12.78
283	152,051	8.18	43	80,149	8.62	20	134,264	12.29	7	122,190	10.17

TABLE 5. Most similar clusters in terms of number of crimes and crime density.

CHD			HDBSCAN			DISTRICT			DBSCAN		
Cluster ID	Num. of crimes	Density	Cluster ID	Num. of crimes	Density	Partition ID	Num. of crimes	Density	Cluster ID	Num. of crimes	Density
80	256,469	80090.48	62	171,400	29061.49	22	249,155	6604.09	4	205,588	16090.5
284	185,769	19143.24	26	165,187	11768.31	16	253,646	2859.60	1	100,646	13524.04
277	466,965	20302.58	28	331,248	18206.78	4	433,057	5668.65	3	223,452	16058.56

550 KM. We extracted all crime reports recorded between January 2010 and December 2019, resulting in 2,110,139 events (an average of 4,059 crimes per week). The dataset totals approximately 321 MB in size.

Figures 17 and 18 present a preliminary temporal analysis of crime trends in Los Angeles. As shown in Figure 17, crime levels remained relatively stable from 2010 to 2014,

followed by a gradual increase until 2018 and a subsequent stabilization through 2019. While a seasonal pattern can be slightly seen in the data, it is less evident than the patterns observed in the Chicago dataset. Figure 18 further supports this observation, displaying a relatively uniform monthly distribution of average crime counts, indicating a stable crime activity trend throughout the year. From the chart,

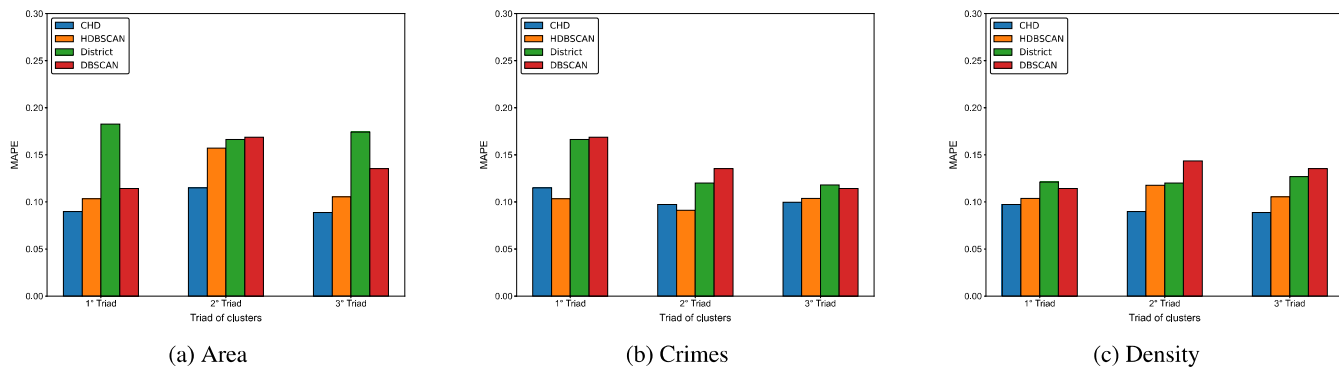


FIGURE 15. Compare first 3 most similar cluster with MAPE, for each category (i.e., Area, Crimes, Density).

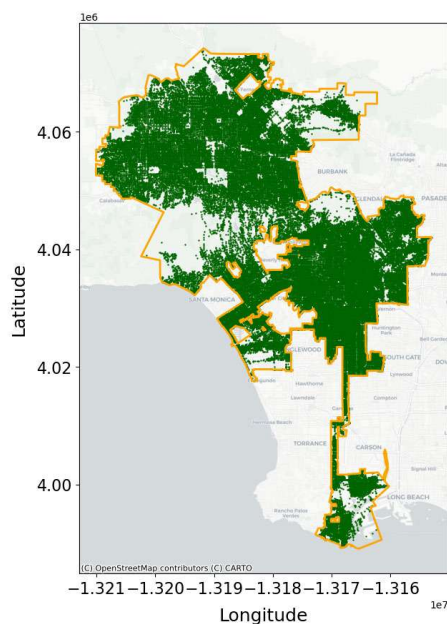


FIGURE 16. Geo-localized crime events in Los Angeles (2010–2019).

we can observe that the highest number of crimes occurs in January (averaging 18,551 crimes, with similar values in May, July, August and October) and the lowest one in February (averaging 15,895 crimes).

The pre-processing step applied to the Chicago dataset was also adopted for the Los Angeles data. Specifically, points outside the administrative boundaries of Los Angeles, which represented less than 0.3% of the entire dataset, were filtered and removed. Furthermore, as done for the Chicago data, the geographic coordinates (i.e., latitude and longitude) were transformed into Cartesian coordinates (i.e., X and Y) in order to allow the use of Euclidean distance in the clustering algorithms.

B. LOS ANGELES: ESTIMATING OPTIMAL PARAMETERS FOR HDBSCAN, CHD, AND DBSCAN

This section outlines the experimental setup and results used to identify the optimal parameter settings for our analysis.

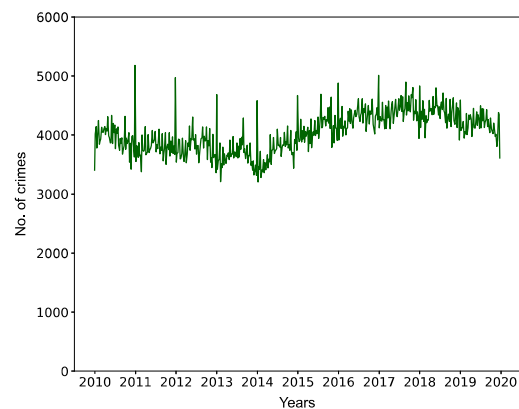


FIGURE 17. Number of crimes vs time (2010–2019).

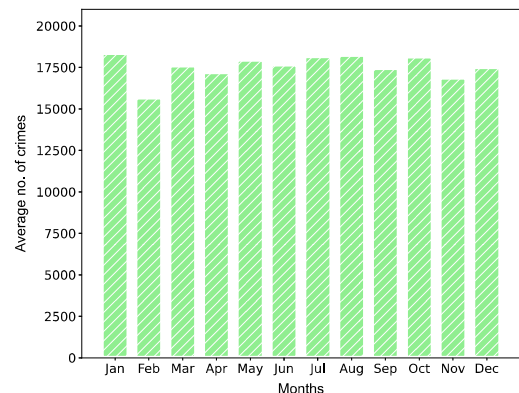


FIGURE 18. Distribution of the number of crimes by month.

As in the Chicago case study, fine-tuning the parameters of the clustering algorithms (i.e., HDBSCAN, CHD, and DBSCAN) is essential to ensure the extraction of meaningful and well-defined crime hotspots, thereby enabling a robust and fair comparison among the four evaluated approaches. Unlike the *Police-Districts* method, which requires no parameterization as it is based on a predefined territorial structure established by law enforcement, the three clustering algorithms depend on specific input parameters. To optimize

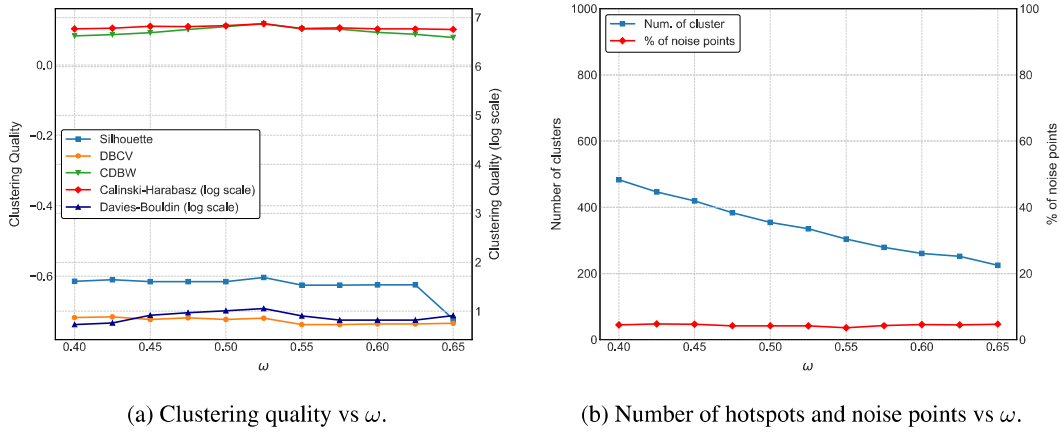


FIGURE 19. Los Angeles crime data. CHD: clustering quality, number of hotspots and noise points.

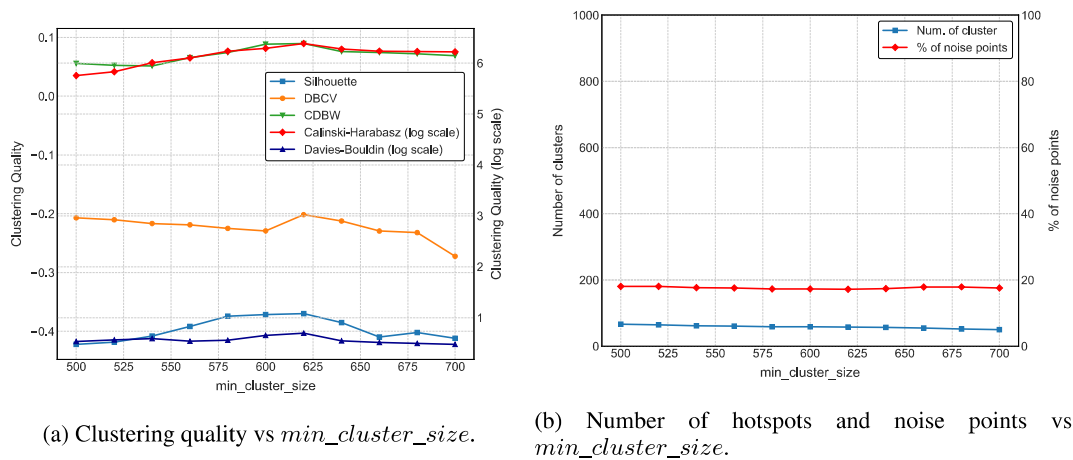


FIGURE 20. Los Angeles crime data. HDBSCAN: clustering quality, number of hotspots and noise points.

these settings, we adopted a parameter-sweeping strategy: each algorithm was executed multiple times across a range of parameter values. The results were then evaluated based on clustering quality indices (Silhouette, DBCV, CDBW, Calinsky-Harabaz, Davies-Bouldin), and the best-performing configurations were selected for subsequent analysis.

The first set of experimental results is illustrated in Figure 19, which evaluates the performance of the CHD algorithm as the parameter ω varies from 0.40 to 0.65. In particular, Figure 19a shows that the clustering quality of the Los Angeles crime data experiences only slight fluctuations across different values of ω . From this plot, we identify the optimal parameter value as $\omega^* = 0.525$. Meanwhile, Figure 19b displays how the number of noise points (red curve) and the number of detected hotspots (blue curve) change with ω . Notably, at the optimal value $\omega^* = 0.525$, the CHD algorithm identifies 88,204 noise points—representing 4.18% of the total data—and detects 335 clusters.

The experimental results for the HDBSCAN algorithm are presented in Figure 20, where clustering performance

is evaluated as the parameter $\min_cluster_size$ varies from 500 to 700. As shown in Figure 20a, the algorithm demonstrates high sensitivity to changes in $\min_cluster_size$, with the best performance achieved at $\min_cluster_size^* = 620$. At this optimal setting, Figure 20b shows a slight downward trend in both the number of detected clusters and the number of noise points as $\min_cluster_size$ increases. This pattern is consistent with observations from the Chicago case study and confirms that larger values for the cluster size threshold typically result in fewer, but more compact, clusters. Specifically, at $\min_cluster_size^* = 620$, HDBSCAN identifies 362,522 noise points, accounting for 17.18% of the total data, and detects 58 clusters.

The results for the DBSCAN algorithm are presented in Figure 21, with the ϵ parameter evaluated across a range from 250 to 350. As shown in Figure 21a, the optimal clustering performance is achieved when $\epsilon = 290$. At this setting, the algorithm identifies 10.55% of the data points as noise and detects a total of 187 clusters (see 21b).

The experimental tests conducted so far have been focused on determining the parameter configurations that yield

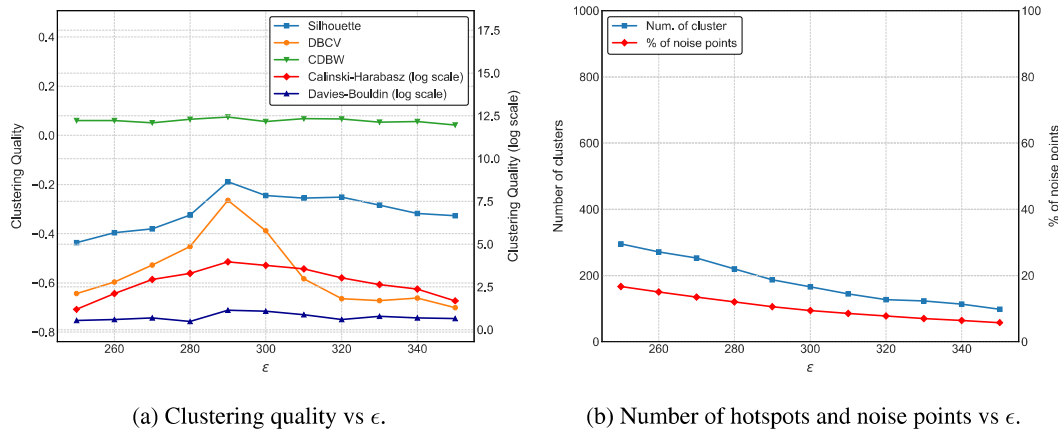


FIGURE 21. Los Angeles crime data. DBSCAN: clustering quality, number of hotspots and noise points.

the highest-quality clusters (i.e., crime hotspots), for each clustering algorithm. Below, we present the results obtained using these optimal settings: for CHD, $\omega = 0.55$, $k = 8$, and $s = 30000$; for HDBSCAN, $min_cluster_size = 620$ and $min_points = 10$; and for DBSCAN, $minPts = 64$ and $\epsilon = 290$.

C. LOS ANGELES: CRIME HOTSPOTS DETECTED BY POLICE-DISTRICTS, HDBSCAN, CHD, AND DBSCAN

We now analyze the crime hotspots identified within the city of Los Angeles. Figure 22 displays the spatial partitions identified by the Police-Districts, HDBSCAN, CHD and DBSCAN approaches. Specifically, Figure 22a shows the predefined territorial divisions into police districts, fixed by law enforcement and publicly accessible through the Los Angeles Open Data Portal,⁸ with the corresponding district boundaries.⁹ On the other side, Figures 22c, 22b, and 22d show the clusters identified by the CHD, HDBSCAN, and DBSCAN algorithms, using the optimal parameter configurations identified in the previous section. We can notice that, as also observed in the Chicago case study, clustering methods identify hotspots with irregular and flexible shapes, unlike the fixed polygonal boundaries of police districts. Table 6 summarizes input parameter values and the main results of each partitioning approach (i.e., number of hotspots/districts and percentage of noise points).

Based on the results shown in Figure 22 and summarized in Table 6, we can make the following observations:

(i) **CHD detects an higher number of crime partitions compared to HDBSCAN, DBSCAN, and Police-Districts.** As presented in Table 2, the Police-Districts approach segments the city into 21 fixed administrative districts. In contrast, the clustering algorithms identify significantly more granular crime hotspots: DBSCAN detects 187, HDBSCAN identifies 58, and

CHD reveals 335 hotspots, respectively. Notably, CHD also achieves the lowest percentage of noise points (4.18%), outperforming the other algorithms in reducing unclustered data points. These findings, consistent with observations from the Chicago case study, demonstrate that clustering-based methods produce significantly more fine-grained spatial partitions. By uncovering smaller, more localized areas of high crime concentration, they enable more precise identification of at-risk neighborhoods. This enhanced granularity supports more targeted crime analysis and informed intervention strategies. Furthermore, CHD stands out for its ability to minimize noise while delivering broader and more detailed coverage of crime hotspots.

(ii) **Police-Districts, DBSCAN, HDBSCAN and CHD detect very different partitions of the city.** As illustrated in Figures 22a, 22d, 22b, and 22c, the partitions/clusters detected by the four methods differ significantly in terms of area coverage, shape, boundaries, and the number of points assigned to each partition/cluster. For instance, HDBSCAN, CHD and DBSCAN identify a large crime hotspot in the mid-east central area of Los Angeles (highlighted in green, red and yellow, respectively), which spans across multiple police districts as defined by Police-Districts. On the contrary, in the central-southernmost part of the city, HDBSCAN, CHD and (partially) DBSCAN detect two main crime hotspots, which are included in only one narrow and long police district. These examples show that the three data-driven approaches detect spatial patterns from crime data, that sometimes are not taken into account by the static subdivision in police districts. Other similar cases can be observed by comparing the four figures.

To provide a comprehensive characterization of the detected clusters with the different approaches in the Los Angeles dataset, we exploit the same figures about the cumulative distributions introduced for the Chicago dataset. For each partitioning approach, Figures 24, 23, and 25 show on the Y axis the percentage of clusters having less than the number of crimes, area (KM²), and crime density (number of crimes/KM²), respectively, as reported on the

⁸Los Angeles Open Data Portal: <https://data.lacity.org>

⁹Los Angeles City Boundaries: https://data.lacity.org/City-Infrastructure-Service-Requests/City-Boundary-of-Los-Angeles/brvb-jr45/about_data

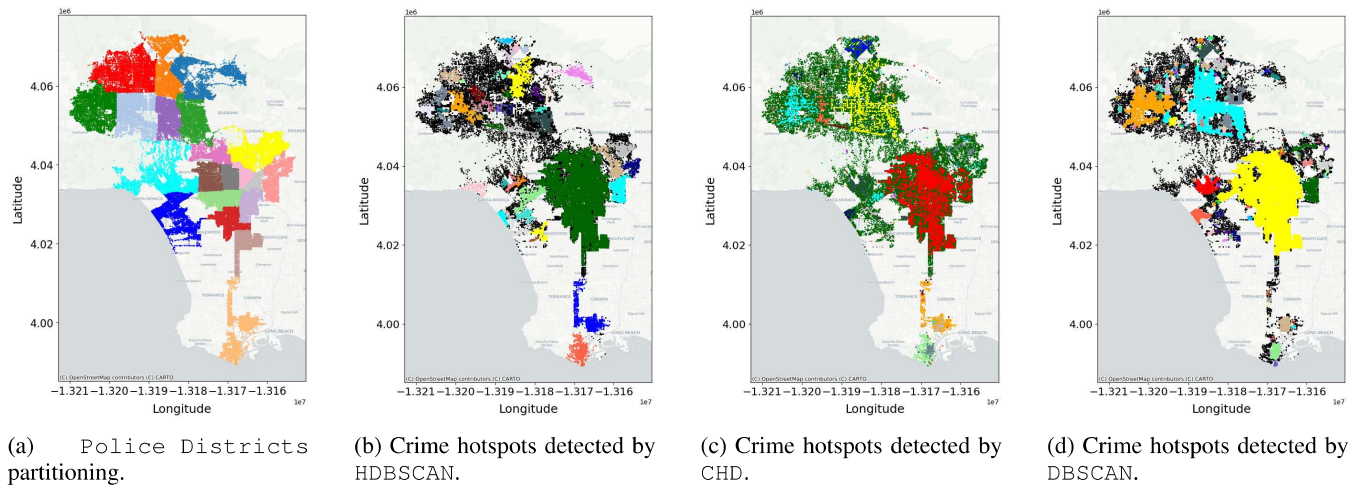


FIGURE 22. Crime partitions detected in Los Angeles.

TABLE 6. Los Angeles. Overview of the clustering results obtained by the different approaches.

Partitioning technique	Input parameters	# hotspots/districts	# noise points
District	—	21	—
HDBSCAN	$\min_cluster_size = 620$, $\min_points = 10$	58	17.18%
CHD	$\omega = 0.525$, $k = 8$, $s = 30000$	335	4.18%
DBSCAN	$\epsilon = 290$, $\min_points = 64$	187	10.55%

X axis. The results lead to similar considerations given for the Chicago dataset. From the figures, we can highlight that, considering the `Police-Districts` partitioning, the number of crimes collected in each pre-given district ranges from 80K to around 150K, and the occurrence rate of such values is more balanced w.r.t. the data-driven partitioning approaches. On the other side, the majority (more than 90%) of hotspots detected by DBSCAN, HDBSCAN and CHD contain less than 50K, 100K and 50K crimes, respectively. Similar considerations can be made for areas (Figure 24) and crime density (Figure 25) of the districts/hotspots. For example, Figure 24 shows that police districts have areas distributed between 10 KM² and 250 KM², with around the 90% of districts smaller than 180 KM², while the majority of hotspots detected by HDBSCAN, and DBSCAN have areas smaller than 25 KM²; on the other side, the majority of hotspots produced by CHD have areas lower than 10 KM². Furthermore, from Figure 11, we can observe similar trends in density highlighted for the Chicago dataset.

D. LOS ANGELES: CRIME FORECASTING RESULTS

In this section, we analyze the impact of different partitioning techniques on crime forecasting performance in Los Angeles. Similarly to the Chicago case study, we assess forecasting accuracy using Mean Absolute Error (MAE) and Mean Absolute Percentage Error (MAPE), as defined in Section V-D. For each cluster generated by the respective

partitioning methods, weekly crime time series have been constructed and modeled using SARIMA with a one-week-ahead forecasting horizon. The models have been trained on crime data from 2010 to 2016, with optimal parameters selected through grid search. Forecasting performance has been then evaluated on the 2017–2019 test set using both MAE and MAPE as evaluation metrics. The quantitative results are shown in Figures 26 and 27, which report the MAPE and MAE metrics for each cluster and for each partitioning technique, ordered by cluster area, cluster size (i.e., number of crimes), and average cluster density. Based on the information provided, we can draw the following considerations.

Forecasting performance tends to improve in hotspots with a higher volume of crime incidents. Figure 26 shows that MAPE values tend to decrease as the number of crimes per cluster (or hotspot) increases. This inverse relationship is clearly observed across all three clustering algorithms, where most clusters contain fewer than 50,000 crime events, confirming the trend previously noted in the Chicago case study. In contrast, this pattern is not evident in the district-based partitioning, primarily because it lacks clusters with fewer than 50,000 crime events, limiting the variability needed to (eventually) observe such a trend.

Multi-Density-based partitioning outperforms district-based partitioning and single-density DBSCAN in forecasting accuracy. HDBSCAN and CHD partitioning approaches outperform district-based and DBSCAN-based partitioning in terms of forecasting accuracy. This is evident from both Figures 26 and 27, where MAPE and MAE achieved by multi-density clustering algorithms are, in general, lower than the other two approaches, particularly for clusters of comparable size. This improved performance, similarly to what we have seen for the city of Chicago, can be attributed to the ability of data-driven partitioning methods to identify spatial areas in Los Angeles that are more homogeneous in crime density, compared to the broader

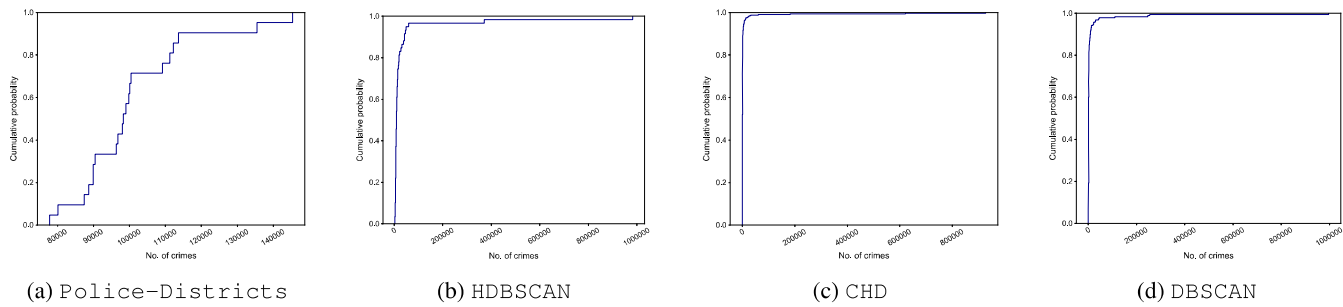


FIGURE 23. Los Angeles: Cumulative distribution of the number of crimes in each district/hotspot, for the three approaches.

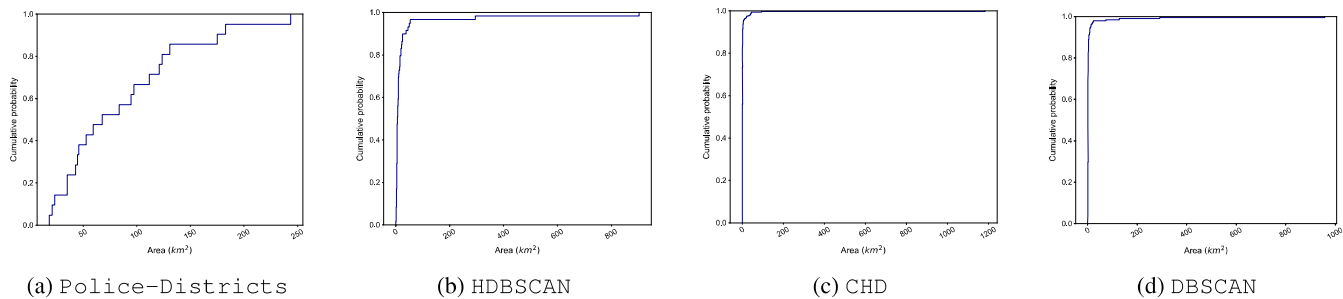


FIGURE 24. Los Angeles: Cumulative distribution of the area in each district/hotspot, for the three approaches.

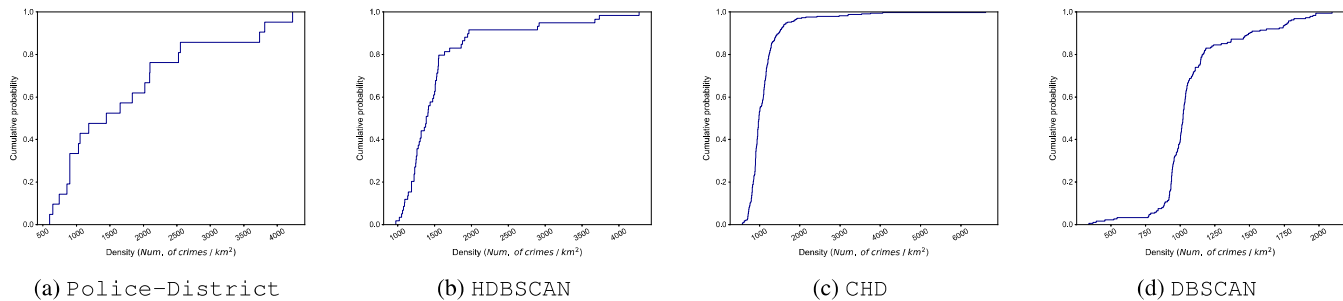


FIGURE 25. Los Angeles: Cumulative distribution in each district/hotspot, for the three approaches.

and administratively defined boundaries of district-based partitioning.

VII. DISCUSSIONS, INSIGHTS, AND OPEN QUESTIONS

This study investigates how different spatial partitioning methods influence the accuracy of crime forecasting, specifically comparing the conventional static division of cities into police districts with data-driven, density-based clustering algorithms for identifying crime hotspots. Through an experimental evaluation using two real-world datasets, i.e. Chicago and Los Angeles, the results clearly show that density-based approaches are more effective in detecting crime hotspots that vary in density, ultimately leading to improved forecasting performance across diverse urban settings. While these findings underscore the potential of clustering-based methods

in enhancing predictive accuracy, several important aspects could need further discussion, including their practical implementation, limitations, integration with existing tools used by law enforcement agencies, etc. To provide a more comprehensive understanding of the topic, we present a set of key questions along with in-depth responses aimed at expanding the discussion and addressing these critical considerations.

Q1. How the accuracy of crime forecasting approaches can be improved ? Crime forecasting accuracy can be enhanced in several ways. First, enriching crime data with external variables (e.g., weather, demographics, events) can enhance the quality of clustering and crime forecasting models. Additionally, applying ensemble or hybrid models, performing regular model updates, can contribute to keep the models up-to-date with dynamic changes of criminal

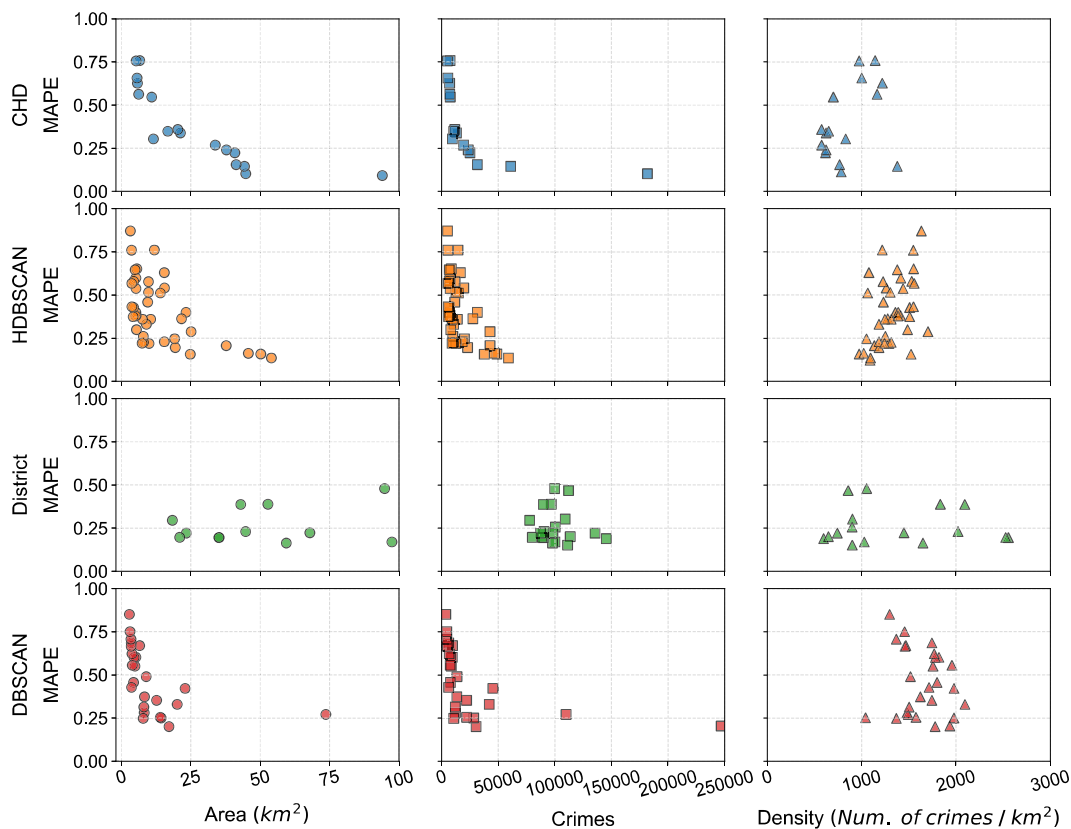


FIGURE 26. Los Angeles: MAPE vs Cluster area, num. of crimes and density.

activities and allow more accurate and robust forecasting outcomes.

Q2. What are the practical implications of the findings, particularly regarding their application in crime prevention strategies and policy-making? What are the advancement of this research findings? The practical implications of using machine learning for crime forecasting, particularly in identifying crime hotspots and building predictive models tailored to each hotspot (as described in this paper), has significant practical implications for both crime prevention and informing policy development. In particular, density-based clustering methods can accurately identify spatial and temporal crime patterns, whose availability can allow law enforcement to proactively allocate resources more strategically, improve patrol routes, and concentrate efforts in high-risk areas. This proactive, data-informed approach enhances operational effectiveness and can contribute to crime reduction through more precise interventions. For urban planners and policymakers, these insights offer a foundation for shaping long-term initiatives, such as targeted infrastructure improvements, community or social programs in persistently high-crime neighborhoods. Furthermore, integrating predictive analytics into existing crime analysis systems encourages transparency and accountability, supporting evidence-based decisions rather than reactive or

intuition-driven approaches. Finally, these findings reinforce the value of adopting adaptive, context-sensitive forecasting tools that respond to the unique dynamics of different urban environments and crime types.

Q3. How do these approaches generalize across different urban settings (e.g., high-density vs. low-density areas) and crime types (e.g., property crime, violent crime)? The crime forecasting approaches described in this paper can be tailored to suit various urban environments and crime types, but their effectiveness depends on how well the models capture the spatial and temporal characteristics of each setting. However, the approaches described here, particularly HDBSCAN and CHD, demonstrate strong adaptability by identifying crime hotspots with varying densities. However, in high-density urban areas, where crime patterns are often more concentrated and data-rich, forecasting models can detect trends more accurately. In contrast, low-density regions can pose challenges due to sparser and more scattered data, often requiring parameter adjustments. Additionally, the type of crime plays a crucial role: for example, property crimes, with their more predictable patterns, are generally easier to forecast than violent crimes, which tend to be less frequent and influenced by complex social and situational factors. To ensure robust generalization across different settings, it is essential to calibrate models to local conditions,

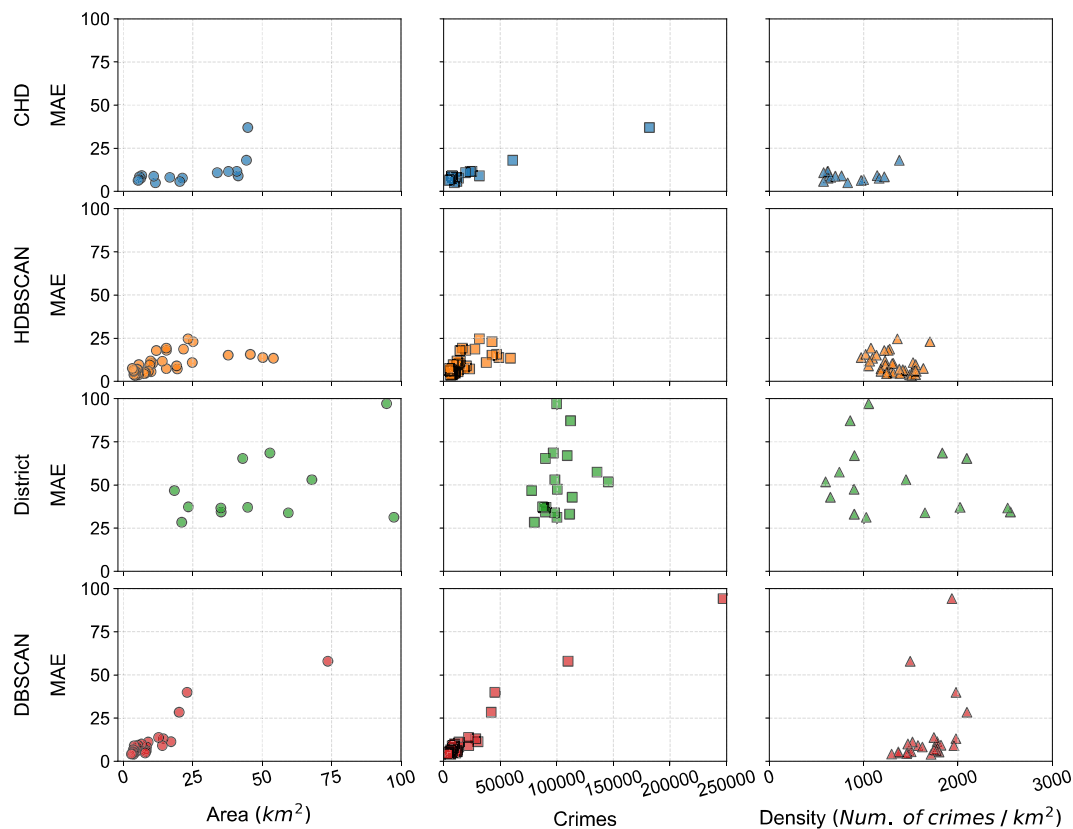


FIGURE 27. Los Angeles: MAE vs Cluster area, num. of crimes and density.

integrate contextual variables, and rigorously validate their performance across diverse environments.

Q4. *How law enforcement agencies or urban planners could integrate these algorithms into existing crime analysis systems?* Law enforcement agencies and urban planners can integrate crime predictive models and crime analysis systems, by incorporating them into GIS platforms or crime mapping tools to dynamically identify and visualize high-crime hotspots and patterns. However, effective integration requires establishing workflows for continuous data collection, regular model retraining and tuning, and systematic interpretation of outputs. Additionally, training personnel in the use and interpretation of these tools is essential to ensure they support data-driven decision-making for strategic resource allocation, urban safety planning, and proactive crime prevention.

Q5. *Can these approaches be used for real-time decision-making or long-term strategic planning? How accurate this algorithm or model for the real-time datasets?* Crime forecasting methods are valuable tools for both real-time decision-making and long-term planning. In real-time scenarios, they assist law enforcement in efficiently deploying resources, swiftly addressing new crime trends, and improving patrol strategies in vulnerable areas. Over the long term, these techniques help identify recurring crime

hotspots, guiding urban planners and policymakers in crafting focused responses such as urban redesign, social initiatives, or policy reforms. By embedding forecasting into both tactical operations and strategic frameworks, agencies can take a more proactive, evidence-based approach to enhancing public safety.

Q6. *The current model relies on static spatial data. What could be the impact of lacking temporal dynamics in capturing short-term crime fluctuations or trends?* Integrating temporal dynamics into spatial clustering involves incorporating the time dimension into the clustering process to better capture evolving patterns in spatio-temporal data, such as crime trends. From a technical viewpoint, the dynamic changing of clusters can be taken into account by weighting the density of each data point through a *decay factor*, which gives less importance to historical information and more weight to recent data.

Q7. *What are the challenges related to integrating socio-economic conditions, population density, or law enforcement policies?* Integrating crime data with socio-economic indicators, law enforcement policies, population density, and related contextual variables offers significant potential for enhancing crime forecasting. However, this integration also presents several key challenges that must be addressed. First, socio-economic data are typically aggregated by

administrative units such as census tracts, which often do not align with dynamically defined crime hotspots. This mismatch necessitates careful spatial interpolation or transformation techniques to ensure meaningful comparisons. Second, aligning these datasets temporally can be difficult, as socio-economic data are usually updated infrequently (often on a semiannual or annual basis) whereas crime data may be available in near real-time. This temporal discrepancy can limit responsiveness and model accuracy. Third, while socio-economic conditions evolve gradually, crime patterns can shift rapidly and unpredictably. Moreover, the relationship between socio-economic factors and crime is complex, often involving non-linear or indirect associations that require advanced modeling approaches, potentially reducing interpretability. These issues complicate data synchronization and increase the complexity of system implementation. Addressing them demands careful preprocessing, temporal and spatial normalization techniques, and cautious interpretation of integrated, multi-source datasets.

Q8. Why are center-based clustering approaches, such as K-means, less suitable than density-based methods for detecting crime hotspots? Center-based clustering algorithms, such as K-means, are not well-suited for spatial analysis due to several limitations. First, these algorithms assume that clusters are globular and of similar size, which does not reflect the irregular shapes and varying densities of real-world spatial phenomena like crime patterns. Second, center-based methods are sensitive to outliers, which can skew the cluster centers and distort spatial groupings. Third, they require the number of clusters to be specified in advance, which is often unknown in spatial data where cluster structure should emerge from the data. Fourth, K-means and similar methods do not deal well with varying densities, making them unsuitable for detecting both dense urban and sparse suburban or rural crime patterns within the same dataset. Finally, these algorithms detect centroid-based clusters and do not consider spatial continuity or connectivity, potentially assigning geographically distant points to the same cluster simply because they have similar distances from the cluster centroid. In contrast, density-based clustering methods are better suited for spatial data, as they naturally handle clusters with arbitrary shapes and densities (in contrast to center-based clustering algorithms), they do not require the predetermination of the number of clusters to be discovered and can detect noise and outliers more effectively. For this reason, the research study reported in this paper focuses exclusively on density-based algorithms, offering a comparative analysis of their performance.

VIII. CONCLUSION

The pervasive diffusion of ICT technologies enables the collection of extensive crime-related data, which, when analyzed using machine learning, can lead to effective spatio-temporal urban events forecasting. In this context, the paper has

presented a comparative study on hotspots detection in urban environments for crime forecasting. Specifically, the manuscript has compared the results achieved when relying on a static division of the city in police districts and when hotspots are built by multi-density clustering algorithms. The experimental evaluation, performed on real-world Chicago crime data, has shown that multi-density clustering algorithms outperform police districts partitioning methods in detecting crime hotspots and achieving accurate crime forecasts.

As practical results/findings of this research study, the paper shows that the application of machine learning in crime forecasting (particularly through density-based clustering and hotspot-specific models) offers significant advantages for both crime prevention and policy development. These techniques enable the identification of precise spatial (i.e., crime hotspots) and temporal (i.e., forecasting models) crime patterns, allowing law enforcement to allocate resources more effectively over the urban territory and target interventions more accurately than traditional police-district-based approaches. For urban planners and policymakers, these insights can support long-term strategies such as infrastructure development, social interventions, and community programs. Moreover, integrating predictive analytics into existing systems promotes evidence-based decision-making, while enhancing transparency and accountability in crime prevention efforts.

In future work, several research directions could be explored. First, we plan to explore the use of hierarchical spatial algorithms, which enable the refinement of clusters by subdividing them when their sizes become excessively large, thereby offering more granularity in hotspot detection and providing more detailed and accurate insights. Second, we plan to analyze the correlations between crime patterns and other urban events, to reveal underlying relationships and interactions. Third, we aim to broaden the scope of spatio-temporal prediction algorithms by applying them to forecast different types of events, thereby expanding their potential applications in urban management and planning. Finally, we plan to integrate socio-economic variables (i.e., income levels, unemployment rates, population density, education, etc.) into the analysis and detection of crime hotspots, with the aim to have a more accurate and deeper understanding of the underlying socio-economic conditions associated with the crime hotspots.

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